TM515

Programming and commissioning safety applications with mapp Safety

Requirements

Training modules	TM210 - Working with Automation Studio
Software	Automation Studio 4.8.2 Automation Runtime 4.82 mapp Safety 5.10 (contains SafeDESIGNER 5.10.1.0) Hardware upgrades ≥ V2.0.0.1 mapp View 5.10
Hardware	X20CPU with at least 256 MB RAM SafeLOGIC/SafeLOGIC-X

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1 Introduction

With this training module, participants will learn how to create a sample project and they will familiarize themselves with the most important functions and configurations in Automation Studio and SafeDESIGNER.
The course provides information about the installation of the software and Automation Help, followed by a detailed explanation of sample application "Saw".
Step by step, the individual sensors and actuators are commissioned and the safety controller and its associated safe I/O modules are parameterized to get "Saw" running.
The training module also contains the documentation for the safety application. A sample project in the safety application is used for illustration.

Figure 1: Technology Package "mapp Safety"

Technology Package "mapp Safety" is used to install and provide the required hardware upgrades for the safety components as well as programming interface SafeDESIGNER including all functionalities for the safety controller and motion control.

> The example code in this documentation is exclusively for implementing the test setup for this training. This example code is not permitted to be adopted for safety applications under any circumstances. Each safe machine application must be subjected to a separate risk analysis and a safety concept must be created.

1.1 Learning objectives

This training module is designed to help participants learn how to use SafeDESIGNER and develop a safety application.

Overview of the most important objectives:

- Participants will learn how individual configuration tasks are divided up between Automation Studio and SafeDESIGNER.
- Participants will learn how to add safe modules to a project and complete the I/O configuration in Automation Studio.
- Participants will learn how to open SafeDESIGNER, log in to a project and use the various features of the SafeDESIGNER interface.
- Participants will learn how to edit the safe module configuration in SafeDESIGNER and set the parameters for the safety equipment being used.
- Participants will learn about the documentation requirements for a safety application and how to create documentation for a project using SafeDESIGNER.
- Participants will learn how to use the integrated help documentation to assist you in working with SafeDESIGNER and the PLCopen Safety library.
- Participants will learn how to access and configure function blocks from the PLCopen Safety library in SafeDESIGNER.
- Participants can create your own function block and configure the inputs and outputs.
- Participants will learn about the various diagnostic tools and procedures for commissioning the safety controller.
- Participants will learn how to transfer a safety application to the SafeLOGIC and SafeLOGIC-X controller and test it with the tools provided.
- Participants can transfer the application to the simulation and test it.
- Participants can create SafeCOMMISSIONING files and activate the machine options via the mapp Safety HMI application.

In order to correctly implement a safety application, it is important that applicable regulations and standards are observed in all phases of the safety application's lifecycle. This training module is limited exclusively to use of Automation Studio, SafeDESIGNER and the mapp Safety Technology Package. This training manual can therefore never replace sound training in safety-related topics.

Safety technology \ Intended use

- Organization of notices
- Qualified personnel

1.2 Safety notices and symbols

Safety notices in this manual are organized as follows:

STOP

Danger: Disregarding these safety guidelines and notices can result in severe injury, death or substantial damage to property.

Warning: Disregarding these safety guidelines and notices can result in severe injury or substantial damage to property.

Caution: Disregarding these safety guidelines and notices can result in injury or damage to property. These instructions are important for avoiding malfunctions.

Additional notices and information in this manual are organized as follows:

Note: Provides important tips and additional information.

Help: Refers to an Automation Help entry that contains further information, data sheets or user's manuals.
Example: Programming \ Variables and data types \ Data types \ Basic data types
Click on the link to open Automation Help.

Example: Shows an example that reinforces what you have learned.

Result: Briefly summarizes the result of a completed task.

Organization of safety notices in external manuals:

This manual contains references to other manuals. How safety notices are organized in external manuals is listed in the respective manual.

Exercise: Task definitions and exercises

The sections highlighted in gray describe exercises and the respective steps to be performed. The exercises are designed to provide a deeper understanding of the information provided.

2 mapp Safety concept and scope of installation

Concept

In the context of mapp Safety, a SafeLOGIC (or SafeLOGIC-X) controller including all associated SafeNODEs, project and option definitions is referred to as **SafeDOMAIN**.

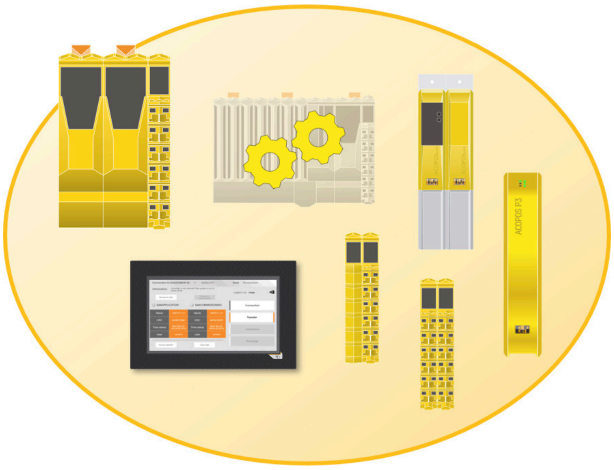


Figure 2: Symbolic representation of SafeDOMAIN

A differentiation is made between the following terms:

- **SafeLOGIC**
  The SafeLOGIC controller is the central processing unit responsible for the cyclic processing of the SafeAPPLICATION as well as configuration and parameter management.
- **SafeNODE**
  The term SafeNODE refers to a safe device that exchanges openSAFETY data with a SafeLOGIC controller. These include both safe I/O modules and safe drives. It is important to note that a SafeNODE must always be uniquely assigned to a SafeDOMAIN. (4 "Configuration in Automation Studio" on page 14)
- **SafeDESIGNER project**
  The SafeDESIGNER project is created using SafeDESIGNER. There, the safety functions are programmed, the I/O channels are linked and the SafeNODEs are configured. (5 "Getting started in SafeDESIGNER" on page 21)
- **SafeAPPLICATION**
  A SafeAPPLICATION is a compiled SafeDESIGNER project. The SafeAPPLICATION is the object that can be transferred to the SafeLOGIC controller. During compilation, each SafeDESIGNER project is automatically entered as a download object in the "Configuration View" underneath SafeAPPLICATION and available for selection on the target system. (6 "Commissioning the safety application" on page 33)
- **SafeOPTION**
  SafeCOMMISSIONING files make it possible to implement safe machine options. (8 "SafeCOMMISSIONING" on page 50)

Safety technology \ mapp Safety \ Concept \ Organization of mapp Safety

Scope of the mapp Safety Technology Package

The mapp Safety Technology Package contains the following contents:

- mapp Safety help documentation
- SafeDESIGNER
- mapp Safety HMI
- Automation Studio hardware upgrades
- B&R tutorials

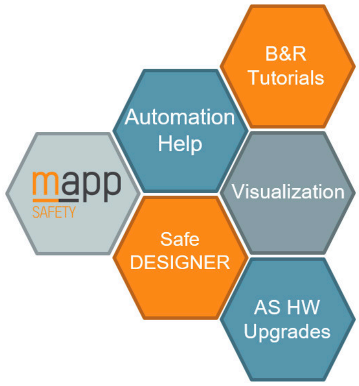


Figure 3: Contents of the mapp Safety Technology Package

2.1 Installation and licensing

Installing the mapp Safety Technology Package

The mapp Safety Technology Package is downloaded and installed via dialog box "Upgrades" in Automation Studio. Alternatively, the Technology Package is available in the Downloads section of the B&R website.

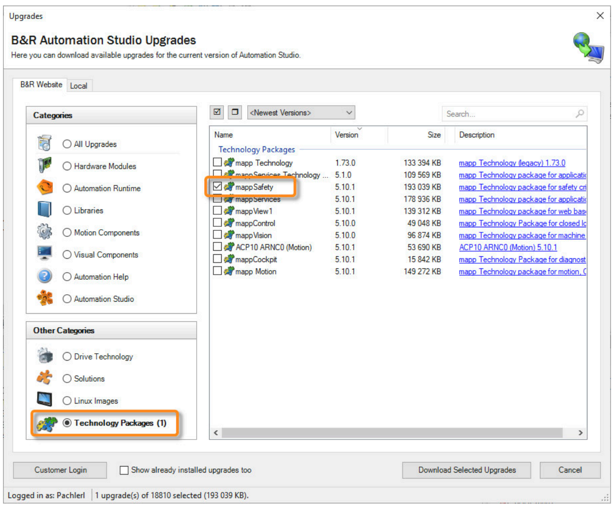


Figure 4: Automation Studio upgrades - mapp Safety Technology Package

Licensing for SafeDESIGNER

For SafeDESIGNER, the licensing mechanism of Automation Studio is used. No separate license is required.

Validity period of mapp Safety license

If the required licenses are not available on the SafeKEY[1], the Technology Guarding system searches for the corresponding licenses. If they are also not available in the Technology Guarding system, a corresponding entry is made in the logbook. This situation is indicated by blinking signals on both the safety controller as well as the standard controller. For information about the required licenses, see Automation Help.

Safety technology \ mapp Safety \ Licensing

2.2 mapp Safety help documentation

Installing the mapp Safety Technology Package integrates all contents required for programming B&R safety technology into Automation Help.
It contains information about the safety hardware, the use of sensors and actuators as well as the engineering of Smart Safe Reaction using mapp Safety and SafeDESIGNER.

1 Storage medium for the SafeLOGIC controller

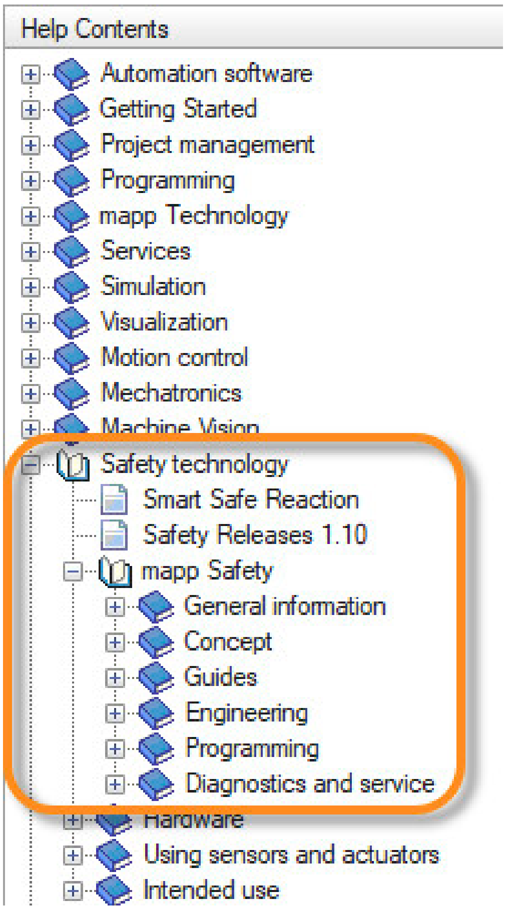


Figure 5: Contents of the mapp Safety help documentation

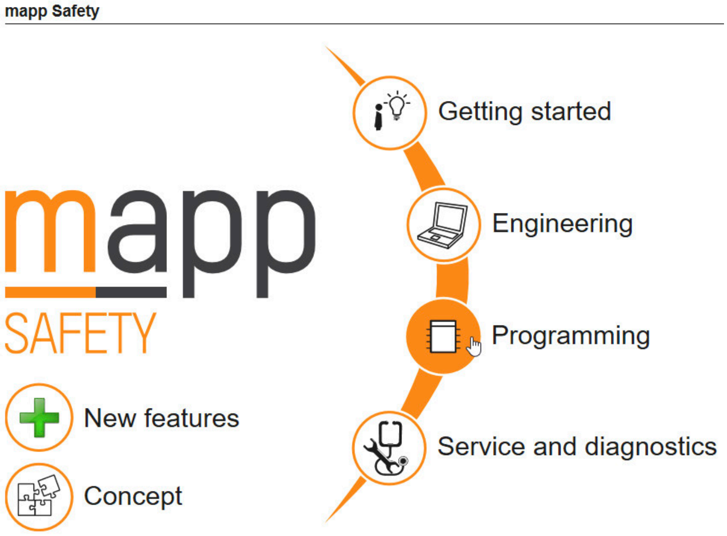


Figure 6: Navigating the start page of the mapp Safety help documentation

Safety technology \ mapp Safety

- Concept
- Getting started
- Engineering
- Programming \ SafeDESIGNER
- Service and diagnostics \ Maintenance scenarios

Safety technology \ Hardware
Safety technology \ Using sensors and actuators
Safety technology \ Intended use

3 Presentation of example project "Saw"

Controlling a saw is an example of a safety application. The image shows a sketch of the hardware structure and the associated safety devices. There are two possible configurations depending on the ETA light system used.
When using a SafeLOGIC controller, it is connected to a CPU via POWERLINK. The safe input and output modules communicate with the SafeLOGIC controller via the X2X interface. Alternatively, this exercise can also be implemented with a SafeLOGIC-X controller.

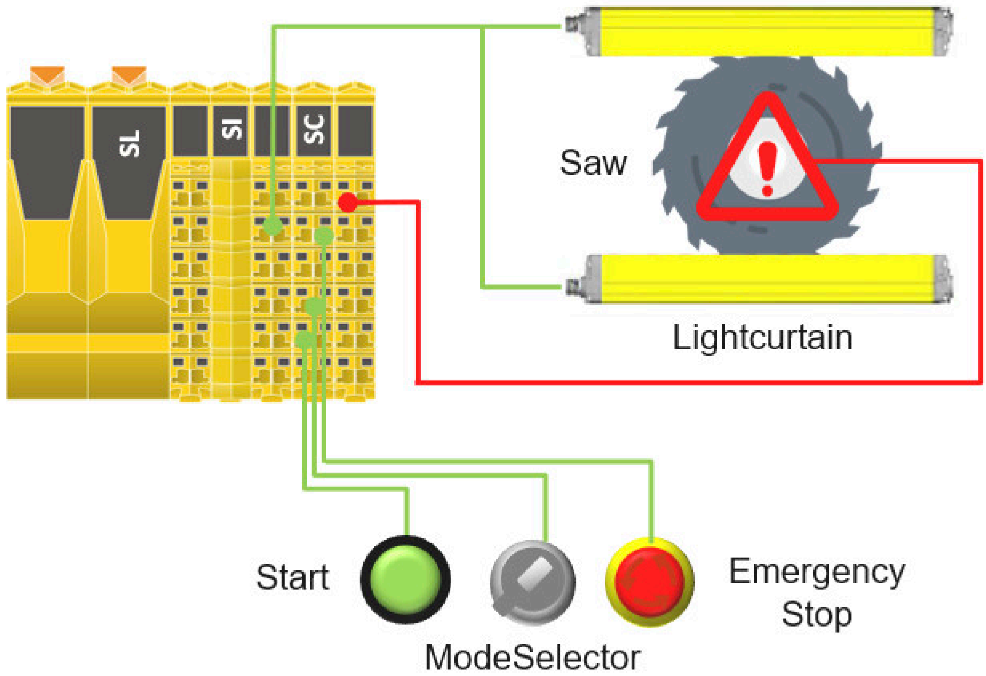


Figure 7: Topology of the hardware used

> The SRS (Safety Requirements Specification) defines the function of the individual safety devices and their interaction. (3.1 "Safety requirements specification (SRS)" on page 11)
> Throughout the entire content of this training module, the individual safety functions in the sample application are implemented and put into operation.

3.1 Safety requirements specification (SRS)

The following safety functions must be implemented in sample project "Saw":

3.1.1 Emergency stop switch function

The following function must be implemented:

- Saw is stopped when triggered.
- No acknowledgment is required after startup.
- Acknowledgment is required via the green button after reseting the emergency stop switch.
- The simultaneity for multi-channel evaluation must be within **200 ms**.

Figure 8: Emergency stop switch

3.1.2 Light curtain function

The following function must be implemented:

- Saw stops when triggered.
- No acknowledgment is required after startup.
- Acknowledgment is required via the green button after an intervention in the light curtain.
- The simultaneity for multi-channel evaluation must be within **300 ms**.
- To support a flexible machine concept, an extension of the safety application is being considered. A safe machine option makes it possible to configure the light curtain as available or not available. SafeCommissioningOption receives the name "S_Disable_Lightcurtain" and can accept values SAFETRUE and SAFEFALSE (8 "SafeCOMMISSIONING" on page 50).

Figure 9: Light curtains

A hard-wired light curtain returns an OSSD signal (Output Signal Switching Device). This OSSD signal must be filtered at the safe input using the switch-off filter.

3.1.3 Mode selector switch function

The following function must be implemented:

- Saw stops when switching.
- No acknowledgment is required after startup.
- Two possible operating modes are used:
  - Manual - Switch position right
  - Automatic - Switch position left
  - Impermissible state - Switch position middle → An error is triggered
- It is not possible to switch the operating modes directly without acknowledgment.
- The maximum time for an invalid state is permitted to be 500 ms.
- The invalid state is acknowledged via the green button.

Figure 10: Mode selector switch

3.1.4 Operating modes and their interaction

Two operating modes are implemented:

Manual operation (switch position right)

- The output is only enabled by the safety application while the green button is pressed and no violation of the safety equipment occurs.
- The light curtain is disabled in this operating mode.
- A violation of the safety equipment must be acknowledged via a rising edge of the green button.

Automatic operation (switch position left)

- With a positive edge of the green button, the output is enabled by the safety application if no violation of the safety equipment (emergency stop, light curtain) occurs.
- The light curtain is active in this operating mode.
- A violation of the safety equipment must be acknowledged via a rising edge of the green button.

Figure 11: Green button

Transferring diagnostic data to the standard application

The diagnostic codes of the individual safe function blocks must be transferred to the standard application via the I/O mapping and passed on to the corresponding variables of data type UINT.

The following variables must be declared:

- diagCode_EStop (DiagCode of "SF_EmergencyStop")
- diagCode_LightCurtain (DiagCode of "SF_ESPE")
- diagCode_ModeSelector (DiagCode of "SF_ModeSelector")

If an error occurs in at least one function block, set one bit to TRUE in Automation Studio.

The following variable must be declared:

- errorInSafetyApplication

4 Configuration in Automation Studio

Automation Studio is the basis for working with mapp Safety. After creating a project, the hardware for the safety functions must first be created. All hardware modules combined in a SafeDOMAIN must have the same SafeDOMAIN ID defined in their configuration.

Module management in Automation Studio includes the following:

- Adding the SafeNODEs (safety controller and safe modules)
- Assigning the SafeNODEs to a SafeDOMAIN
- Determining the project name of the safety application
- Defining the SafeKEY password
- Assigning a user role to access the mapp Safety HMI application
- Accessing disclosed I/O data of the safety components
- Exchanging data between the default CPU and safety controller via communication channels

Each SafeNODE that is added has a SafeNODE ID. This ID must be assigned to a SafeDOMAIN. At least one SafeLOGIC or SafeLOGIC-X controller must be configured for the SafeDOMAIN ID.

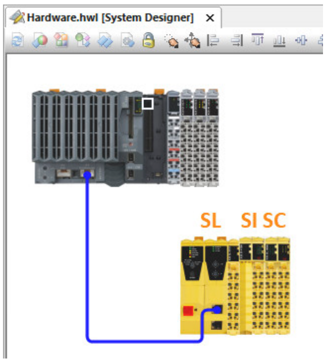


Figure 12: Example hardware configuration

	SafeDOMAIN ID	SafeNODE ID
SL[2]	1	1
SI[3]	1	2
SC[4]	1	3

Table 1: Assigning the SafeNODE ID to a SafeDOMAIN ID

4.1 Preparing an Automation Studio project

To use the mapp Safety HMI application, the **OPC UA system** must be enabled in the CPU configuration.
In addition, the **mapp Safety** and **mapp View** versions used must be **configured** in the Automation Studio project under "Project \ Change runtime version".
Role "SafeDEV" and user "Safety" are created with any password in section "Access and security \ UserRole system" of the Configuration View.

2 SafeLOGIC

3 Safe input module

4 Safe digital mixed module

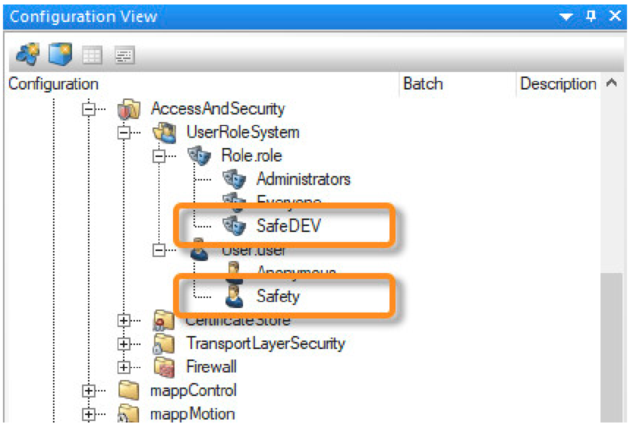


Figure 13: Creating a role and a user

Safety technology \ mapp Safety \ Getting started \ Preparing the Automation Studio project

4.2 Adding a safety controller

The SafeLOGIC controller is the central processing unit responsible for the cyclic processing of the SafeAPPLICATION as well as configuration and parameter management. The SafeLOGIC controller is added in Automation Studio to a POWERLINK interface and the SafeLOGIC-X controller is added to an X2X interface.

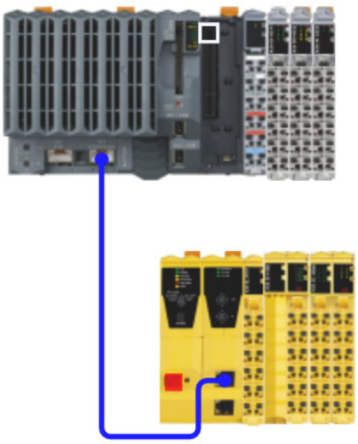


Figure 14: Hardware design with an X20 controller and a SafeLOGIC controller

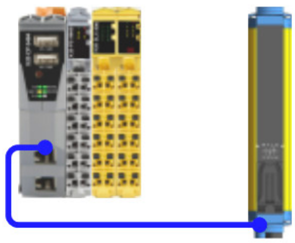


Figure 15: Hardware design with an X20 controller, a SafeLOGIC-X controller and an openSAFETY light curtain

The safety controller is assigned to a POWERLINK interface (SafeLOGIC) or X2X interface (SafeLOGIC-X) via drag-and-drop from the Hardware Catalog.

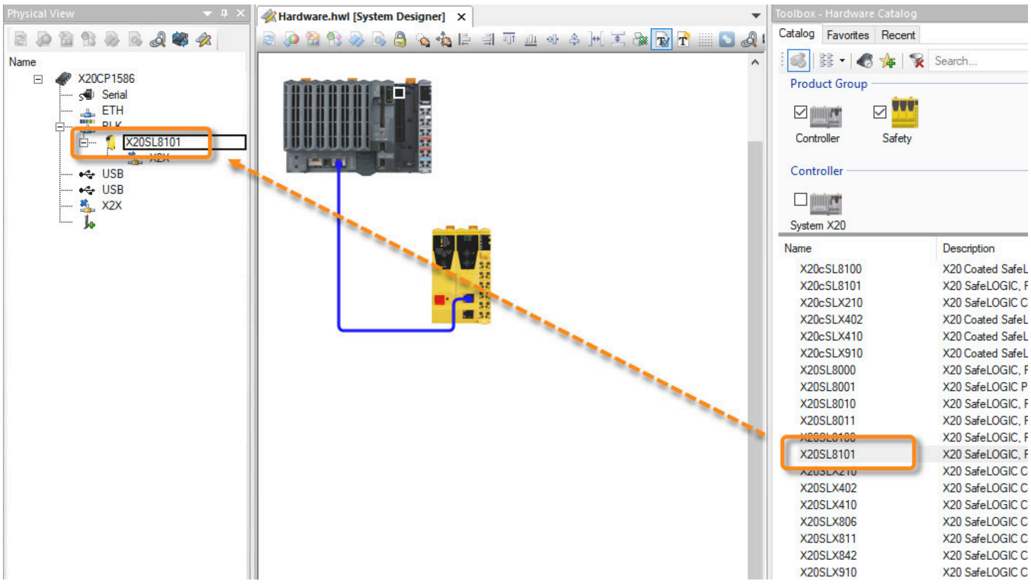


Figure 16: Adding the safety controller from the Hardware Catalog

Setting the POWERLINK node number

The POWERLINK node number of the SafeLOGIC controller can be configured in the Physical View.

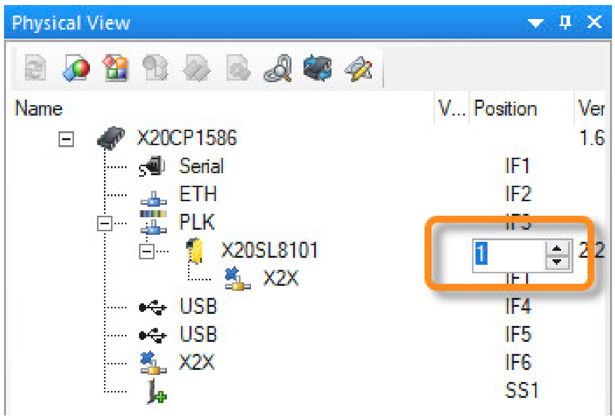


Figure 17: Setting the POWERLINK node number

4.3 Configuring mapp Safety

When using mapp Safety, it is necessary to add configuration files "General" and "SafeDOMAIN" from the Toolbox to the Configuration View and configure them.

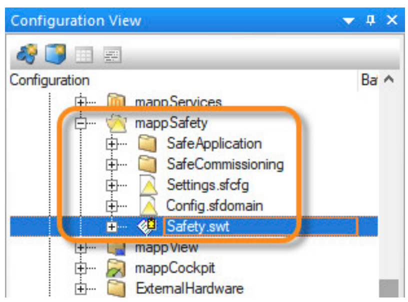
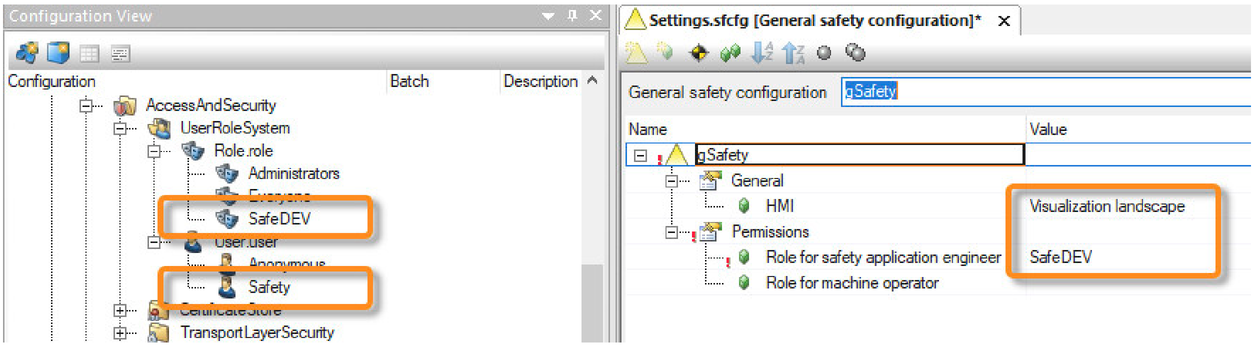


Figure 18: Folder "mapp Safety"

General configuration

The settings in file "Settings.sfcfg" apply to all SafeDOMAINs. The settings are used to manage which mapp Safety HMI application is displayed and which roles have access to it. (6.3 "Download via the mapp Safety HMI application" on page 34)



Configuring the SafeDOMAIN

File "Config.sfdomain" is used to define the SafeKEY password and assign a name to the SafeDESIGNER project.

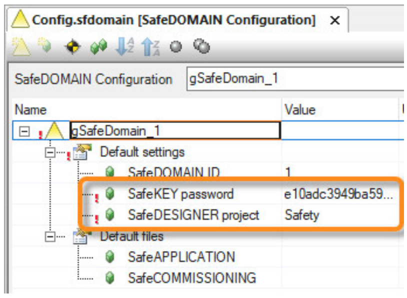


Figure 19: SafeDOMAIN configuration

Safety technology \ mapp Safety \

- Concept \ Configuration of mapp Safety
- Getting started \ Configuring mapp Safety

4.4 Adding SafeIO modules

The SafeIO modules can be added directly to the X2X Link interface of a controller or a POWERLINK bus controller. SafeIO modules can be combined with standard I/O modules as needed.
The SafeNODE ID is used to identify the individual SafeIO modules in a SafeDOMAIN.
The safe I/O modules can be added from the Hardware Catalog at the desired position via drag-and-drop.

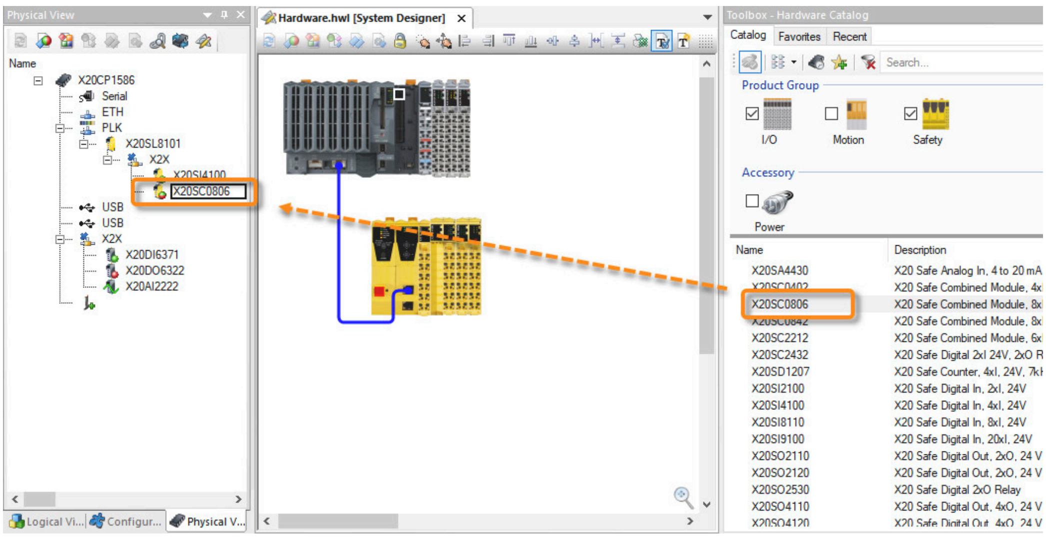


Figure 20: Adding SafeIO modules via drag-and-drop

Module configuration

The shortcut menu of the safe I/O module is used to open the module configuration. This can be used to configure different settings depending on the module type.

Safety technology \
- Hardware
- Using sensors and actuators

4.4.1 Configuring the enabling principle

The following 2 settings can be used to configure the enabling principle:

- "**direct**" (default value): Output channel is visible in the I/O mapping in Automation Studio
  The "direct" mode makes it possible to use the safe output like a default output from the point of view of the standard application. This means that as long as there is no safety request, the output is switched on and off via the standard application.
- "**via SafeLOGIC**": Output channel is hidden in the I/O mapping
  The "via SafeLOGIC" mode is used to control the output via the safety controller only.

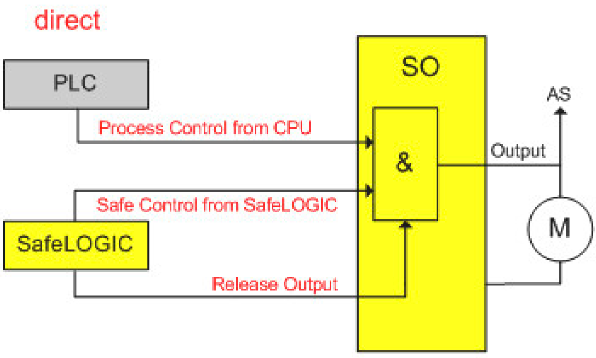


Figure 21: Enabling principle: Mode "direct"

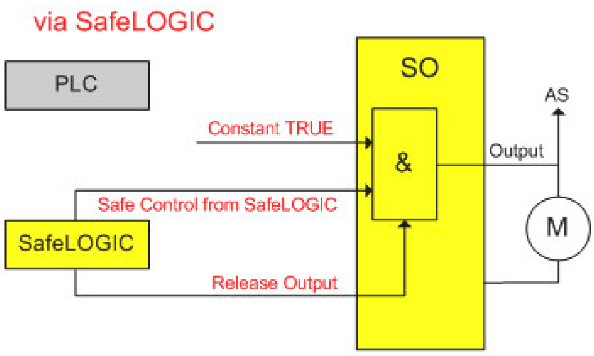


Figure 22: Enabling principle: Mode "via SafeLOGIC"

The enabling principle can be configured individually for each channel of a safe output module. Depending on the selection, the output channel is shown or hidden in the I/O mapping in Automation Studio.

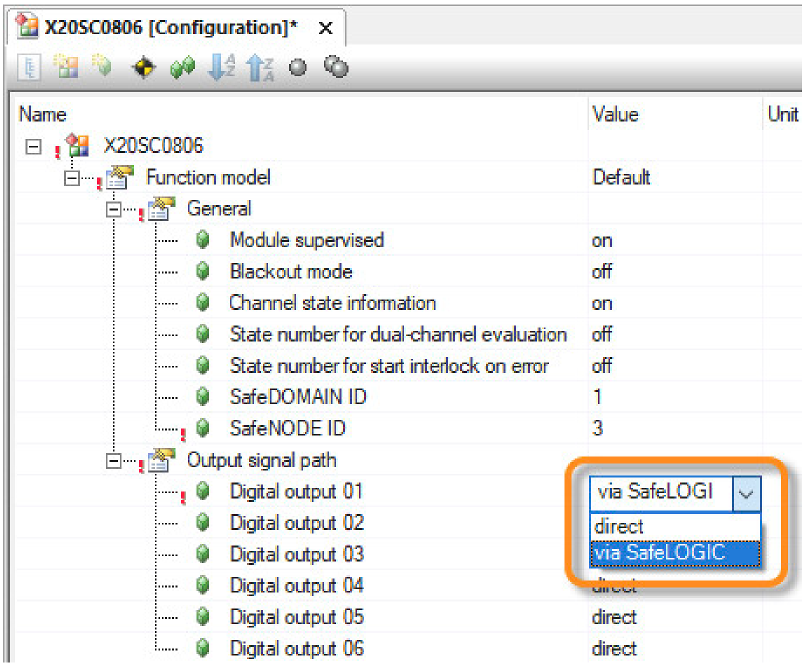


Figure 23: Settings for the enabling principle in Automation Studio

Safety technology \ mapp Safety \ Engineering \ SafeIO \ Safe digital outputs \ Enabling principle

4.4.2 Status of "start interlock on error" on the safe output module

The I/O configuration of safe output modules can be used to define the visibility of the channel status information and the state numbers of "start interlock on error".
For detailed information about the states of "start interlock on error" and the associated state diagram, see Automation Help.

Channel "FBOutputState" can be used to read the state number of "start interlock on error". To make this input visible in the I/O mapping, it must be enabled in the module configuration.

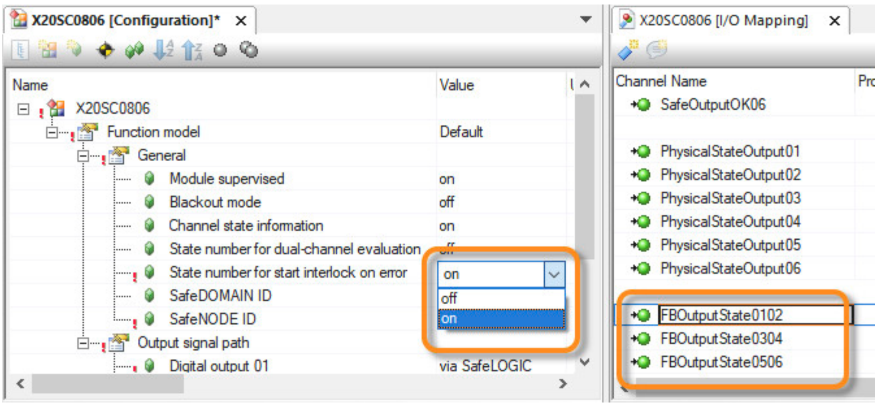


Figure 24: Settings for "FBOutputState"

Safety engineering \ mapp Safety \ Engineering \ SafeIO \

- Restart behavior
- Safe digital outputs \ "start interlock on error" state diagram

4.4.3 Status inputs for multi-channel evaluation

Safe digital input modules are equipped with integrated multi-channel evaluation. The multi-channel evaluation state in the standard application can be evaluated via the I/O mapping in Automation Studio.

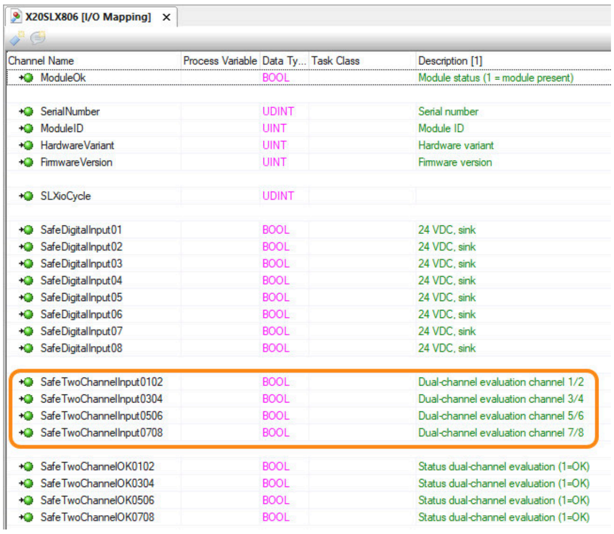


Figure 25: I/O configuration of a SafeLOGIC-X controller - Status of multi-channel evaluation

Depending on whether the equivalence or antivalence between two safe inputs has been defined in SafeDESIGNER, a status is created in the safe input module and made available in the I/O mapping in Automation Studio.

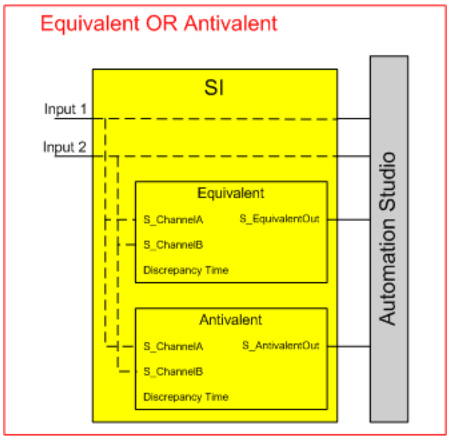


Figure 26: Multi-channel evaluation: Equivalence - Antivalence

SafeDESIGNER provides dual-channel evaluation for each input pair. The setting of the equivalence or antivalence and the discrepancy time must be defined in the device configuration.

Safety technology \ mapp Safety \ Engineering \ SafeIO \ Safe digital inputs \ PLCopen state diagrams "Antivalent" / "Equivalent"
Hardware \ X20 system \ X20 modules \ Digital input modules \ X20(c)SIx10x \ Register description X20SIx1x0 \ Channel list

Exercise: Create a new Automation Studio project, add modules and edit the configuration

The objective of this exercise is to create a new Automation Studio project and to plan and configure the hardware. This project is then transferred to the controller.

1) Create an Automation Studio project.

2) Add CPU and I/O

3) Configure the SafeLOGIC controller

4) Create new role "SafeDEV" and new user "Safety"

5) Insert and configure the Settings.sfcfg and Config.sfdomain file in Config View

6) Configure SafeIOs

7) Configure safe output "via SafeLOGIC"

8) Establish a connection to the CPU

9) Compile and transfer the project.

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Safety technology \ mapp Safety \ Guides \ Getting started

5 Getting started in SafeDESIGNER

SafeDESIGNER can be used to create the safety application, which is cyclically processed by the safety controller. The SafeLOGIC controller and the SafeIOs are also configured in SafeDESIGNER. For this, only the safety-relevant hardware from the Automation Studio configuration is applied.
The safety application can be opened via the shortcut menu of the SafePLC.
Before SafeDESIGNER can be opened, the project name for a SafeDOMAIN must be defined in file "config.sfdomain". (4.3 "Configuring mapp Safety" on page 16)

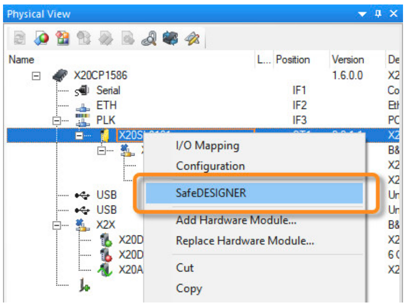


Figure 27: Open SafeDESIGNER.

Safety technology \ mapp Safety \ Getting started \ Creating a SafeAPPLICATION \ Launching SafeDESIGNER for the first time

5.1 Password protection

When a SafeDESIGNER project is opened for the first time, a password must be entered. Passwords must contain at least 6 alphanumeric characters.
This password must be entered each time you wish to access the SafeDESIGNER project.

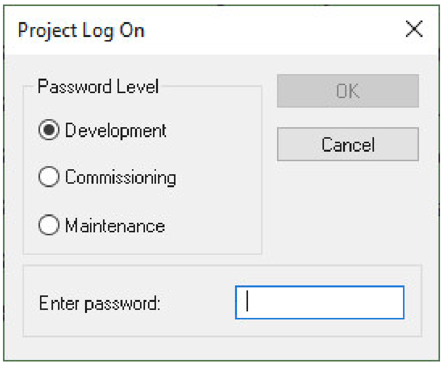


Figure 28: Entering the password in SafeDESIGNER

SafeDESIGNER distinguishes between 3 user levels.

- Development:
  All rights
- Commissioning:
  Open project, change commissioning parameters and download
- Maintenance or "Cancel":
  Open project and download - no password required

After successfully logging in, SafeDESIGNER functions are available based on the user level.

The passwords are set as project specific. If a new project is created, new passwords must be set.

Safety technology \ mapp Safety \ Programming \ SafeDESIGNER \ Introduction \ Password protection for projects and the safety controller

5.2 SafeDESIGNER layout

SafeDESIGNER has multiple toolbars, configuration editors and editing windows. For more information about using SafeDESIGNER, see Automation Help.

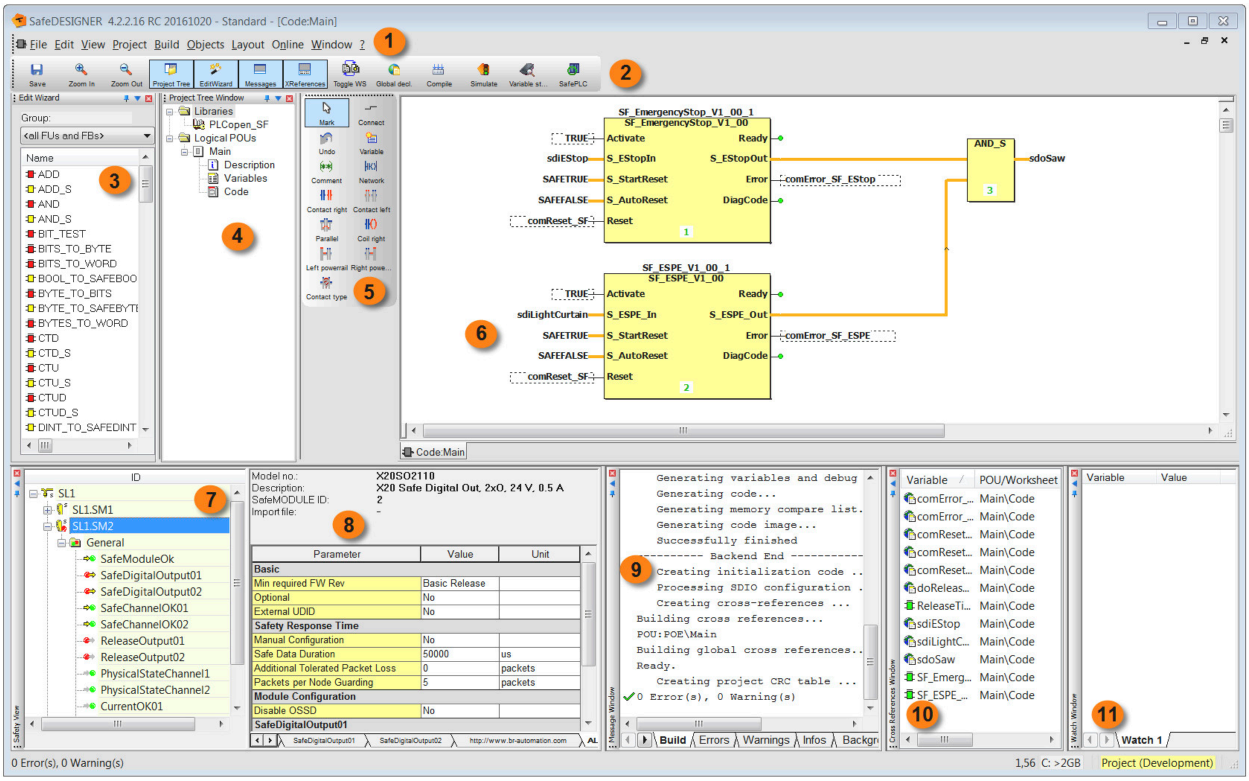


Figure 29: The SafeDESIGNER workspace

Safety technology \ mapp Safety \ Programming \ SafeDESIGNER \

1) User interface \ Menu bar
2) User interface \ Toolbars
3) User Interface \ Edit wizard
4) User interface \ Project tree - Overview
5) User interface \ Toolbars
6) User interface \ Workspace
7) Programming a project \ Device configuration
8) Programming a project \ Device configuration
9) User interface \ Message window
10) User interface \ Cross-references window
11) User interface \ Watch window

5.3 Editor functions

The main window in SafeDESIGNER features the Edit wizard, the project tree and the workspace.

- Edit wizard: Used to add functions and function modules in code worksheets
- Project tree: Used to manage and edit project structure
- Workspace: Used to open programs, variable declarations and descriptions

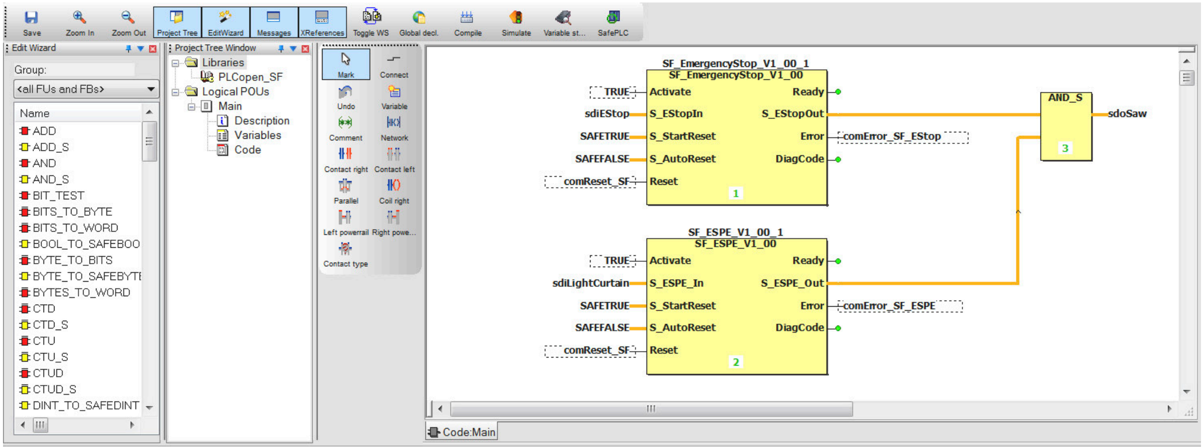


Figure 30: Programming interface in SafeDESIGNER

Safety technology \ mapp Safety \ Programming \ SafeDESIGNER \

- User interface
- EC 61131 implementation \ Data types
- Developing a project

Limited variability language (LVL)

SafeDESIGNER uses the limited variability languages (LVL) Function Block Diagram and Ladder Diagram. This protects the user from errors, such as the creation of endless loops, memory corruption due to corrupted pointers or invalid access to global variables.
Software development is also simplified by ISO 13849-1.

5.3.1 Edit wizard

The Edit wizard can be used to add available function blocks to the worksheet.
This view can be grouped by function groups and libraries.

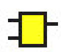
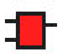
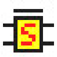
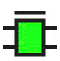
	Safe function / function block per IEC 61131-3
	Non-safe function / function block per IEC 61131-3
	PLCopen Safety function block
	Self-created function / function block

Table 2: Color legend of the function blocks in SafeDESIGNER

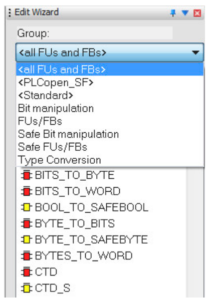


Figure 31: Edit wizard

Safety technology \ mapp Safety \ Programming \ SafeDESIGNER \ User Interface \ Edit wizard

5.3.2 Programs and libraries

The programming languages available for the main code are Ladder Diagram and Function Block Diagram. These can be combined within the worksheets. It is possible to structure the safety application by using multiple worksheets. The user can also create his own function blocks and reuse them within the program. The function blocks are implemented either using Ladder Diagram, Function Block Diagram or Structured Text.

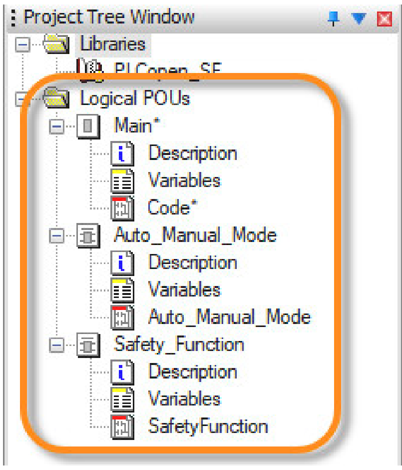


Figure 32: Project tree

Safety technology \ mapp Safety \ Programming \ SafeDESIGNER \ EC 61131 implementation \ Programming languages

5.3.3 Adding functions and function blocks

Functions and function blocks can be added either via drag-and-drop or by double-clicking on the respective function or function block.

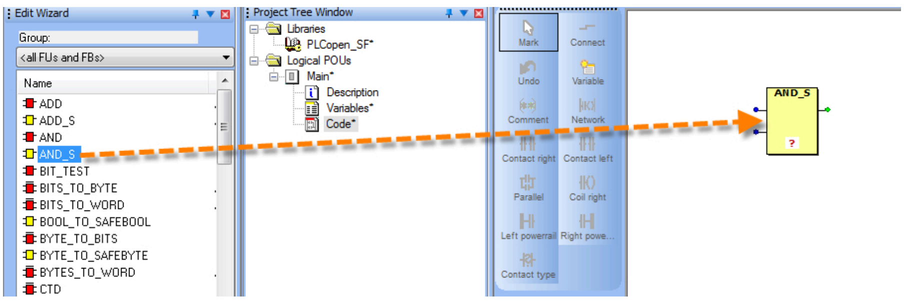


Figure 33: Adding a function in worksheet via drag-and-drop

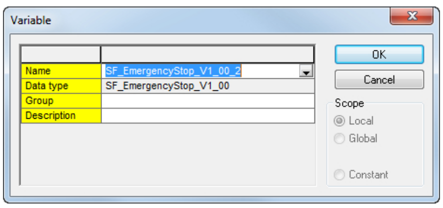


Figure 34: Declaring an instance variable for a function block

Adding a function block opens a dialog box that is used to create an instance variable with the corresponding data type.

Safety technology \ mapp Safety \ Programming \ SafeDESIGNER \

- User Interface \ Edit wizard
- Developing a project \ Developing visual code

5.3.4 Creating a variable

There are precise definitions for the use and scope of local and global variables as well as data types when programming with programming languages with limited variability language (see 5.3 "Editor functions" on page 22).
There are two options for adding a new variable. It is either added via the "Variable" button in the vertical toolbar or by double-clicking on the input or output of a function or function block.

Figure 35: Adding a new variable

A wizard opens to set the variable name, data type and the scope for the variable.

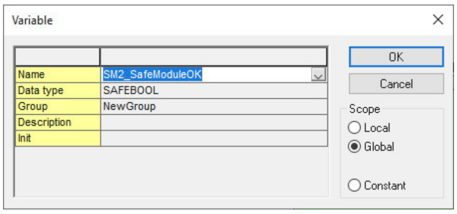


Figure 36: Wizard for the new variable

The following scopes are available for selection:

- Local: For the current worksheet
- Global: Only for communication channels and I/O channels or for the use in multiple worksheets
- Constant: Variable with a constant value

Local variable and constants

Local variables and constants are only used in the current worksheet.
The following button is used to switch between the code view and the variable declaration of the local variables and constants:

Figure 37: Displaying local variables

Global variable

Global variables are exclusively used for communication channels and I/O channels or for the use in several worksheets.
The following button is used to open the variable declaration of the global variables:

Figure 38: Displaying global variables

Safe and non-safe data types

Data types define the properties for the values of a variable. They define the initial value, range of possible values and number of bits. SafeDESIGNER contains elementary data types defined in IEC 61131-3 and special safety-relevant data types.

Safe data type	Non-safe data type
SAFEBOOL	BOOL

Table 3: Safe and non-safe data types

Safe data type	Non-safe data type
SAFEINT	INT
SAFEDINT	DINT
SAFETIME	TIME
SAFEBYTE	BYTE
SAFEWORD	WORD
SAFEDWORD	DWORD

Table 3: Safe and non-safe data types

Variable declaration

The global or local variables are displayed in the variable declaration depending on the selection.

	Name	Data type	Description	Terminal
1	ModuleOK			
2	SM2_SafeModuleOK	SAFEBOOL		SL1.SM2.SafeModuleOK
3	SM3_SafeModuleOK	SAFEBOOL		SL1.SM3.SafeModuleOK
4	SafeCOMMISSIONING			
5	Disable_Lightcurtain	SAFEBOOL		SL1.SM1.SafeCommissioningOptionBIT0...
6	SafeInput			
7	SI_EStop	SAFEBOOL		SL1.SM3.SafeTwoChannelInput0102
8	SI_Lightcurtain	SAFEBOOL		SL1.SM2.SafeTwoChannelInput0102
9	SI_AutoMode	SAFEBOOL		SL1.SM3.SafeDigitalInput03
10	SI_ManualMode	SAFEBOOL		SL1.SM3.SafeDigitalInput04
11	SI_Start	SAFEBOOL		SL1.SM3.SafeDigitalInput05
12	SafeOutput			
13	SO_Saw	SAFEBOOL		SL1.SM3.SafeDigitalOutput01
14	ReleaseOutput			
15	RO_SC_Module	BOOL		SL1.SM3.ReleaseOutput
16	Communication			
17	comDiagCode_EStop	WORD		SL1.SM1.UINT0001
18	comDiagCode_Lightcurtain	WORD		SL1.SM1.UINT0002
19	comDiagCode_ModeSelector	WORD		SL1.SM1.UINT0003

Figure 39: Global variable declaration

Constants can be added via the entries from DATA TYPE#VALUE. Example: "SAFETIME#10s" for a time constant with a duration of 10 seconds.

Safety technology \ mapp Safety \ Programming \ SafeDESIGNER \

- User interface \ Variables worksheets
- User interface \ Dialog boxes for code editing \ Dialog box 'Variable"
- User interface \ Visual code objects \ Variables
- Developing a project \ Developing visual code \ Adding and editing constants (literals)

5.3.5 Linking a function / function block

There are two ways to link a variable to an input or output of a function or function block.
The variable can be set directly to the input or output by moving it.

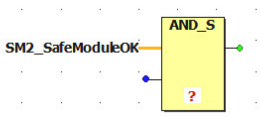


Figure 40: Variable linked to function input

Button "Connect" can also be used to connect a variable with a function / function block or two functions / function blocks.

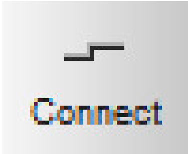


Figure 41: Button "Connect"

A connecting line between two contacts is drawn via clicking and dragging.

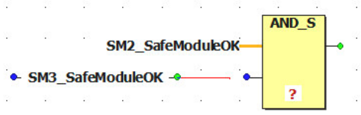


Figure 42: Drawing a connecting line

Safety technology \ mapp Safety \ Programming \ SafeDESIGNER \ User interface

- Keyboard shortcuts
- Visual code editor (general description)
  - Handling objects in the code editor
  - Connecting objects in the code editor

5.3.6 Adding a Ladder Diagram network

A network can be added via a button in the vertical toolbar.

Figure 43: Adding Ladder Diagram network

A network is added at the current cursor position in the worksheet.

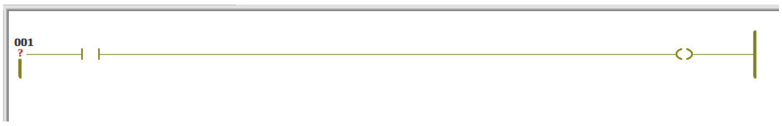


Figure 44: New network in worksheet

Local or global variables must now be connected to the network elements. A double click on the element opens a wizard for selecting a variable.

Adding function blocks to networks

It is also possible to create networks with a function / function block.

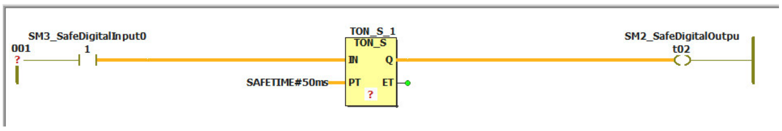


Figure 45: Network with function block

Safety technology \ mapp Safety \ Programming \ SafeDESIGNER \ User interface \ Graphical code objects

- Contact and coil
- Function and function blocks
- Variable

5.3.7 Adding a comment in the worksheet

Button "Comment" can be used to add a comment at the current cursor position. A window will open to enter the text and configure various formatting settings.

Figure 46: Button to add a comment

After the window has been acknowledged, the text is added at the current cursor position.

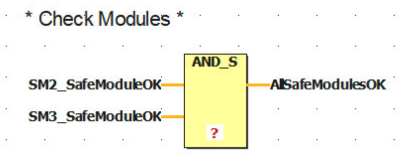


Figure 47: Comment in the worksheet

Safety technology \ mapp Safety \ Programming \ SafeDESIGNER \ User interface \ Visual code editor (general description) \ Adding comments - Dialog box "Comment"

5.4 Linking I/O channels

The I/O mapping in the Safety View serves as an interface between the channels of the SafeNODEs and SafeDESIGNER. Global variables are connected to the safe inputs and outputs via the global variable declaration in SafeDESIGNER.

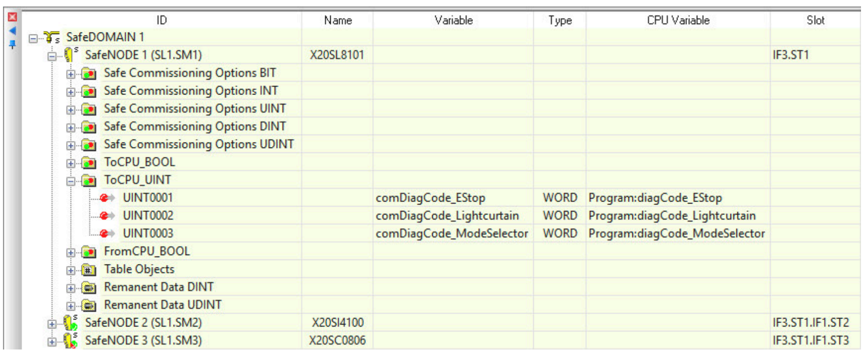


Figure 48: Safety View

Column "Slot" displays the physical position of the safe modules. This is for information only and cannot be changed.
Column "Variable" contains the name of the I/O data point in the safety application.
Column "CPU variable" contains the name that was configured for the I/O data point in Automation Studio.

All channels with a yellow arrow are safety-related and must reference to variables with safe data types.

Safety technology \ mapp Safety \ Programming \ SafeDESIGNER \ Developing a project \ Connecting/Disconnecting process data elements and global I/O variables

5.4.1 Channels of the SafeLOGIC controller

Configurable communication channels and SafeCOMMISSIONING channels are available for the SafeLOGIC controller.
The communication channels can be used, for example, to transmit the DiagCodes of the PLCopen blocks to the standard application. In addition, the channels of the SafeLOGIC controller can be used to transmit a non-safe signal of an acknowledgment button from the standard application to the safety application.
For more information about SafeCOMMISSIONING, see chapter 8 "SafeCOMMISSIONING" on page 50.

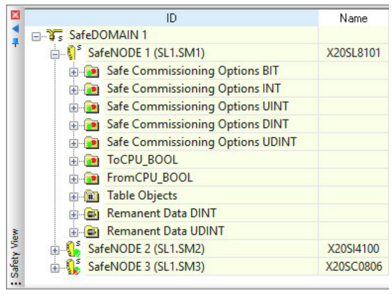


Figure 49: I/O mapping - SafeLOGIC

Safety technology \ mapp Safety \ Hardware \ X20 system \ Module overview: Alphabetical

5.4.2 Safe input and output modules

The individual input and output channels and channels for status information are available for connecting global variables on the safe I/O modules.

In addition, it is possible to use multi-channel evaluation for the safe inputs (4.4.3 "Status inputs for multi-channel evaluation" on page 19).

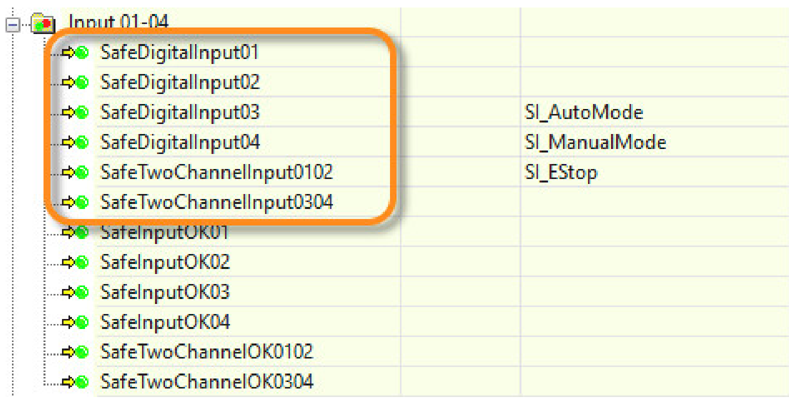


Figure 50: Input channels

Any safe output module is equipped with an internal "start interlock on error" to switch on the channel after an error on the module and/or network. The error is acknowledged via a rising edge on channel "ReleaseOutput".

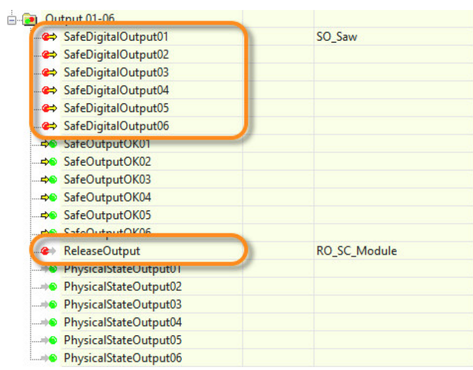


Figure 51: Output channels

If no data points have been linked in SafeDESIGNER by a safe module, the SE LED on the safe module blinks (9.2.1 "Status elements" on page 62).

Safety technology \ mapp Safety \ Hardware \ X20 system \ Module overview: Alphabetical

5.4.3 Linking an I/O channel to a new variable

To define an I/O channel as a new variable, the channel is first selected in the Safety View and then dragged and dropped onto the worksheet.
The declaration dialog box for a new global variable appears. After the variable name has been specified, the dialog box can be confirmed and the new variable can be placed on the worksheet as needed. The new variable name is now assigned to the corresponding channel in the Safety View.

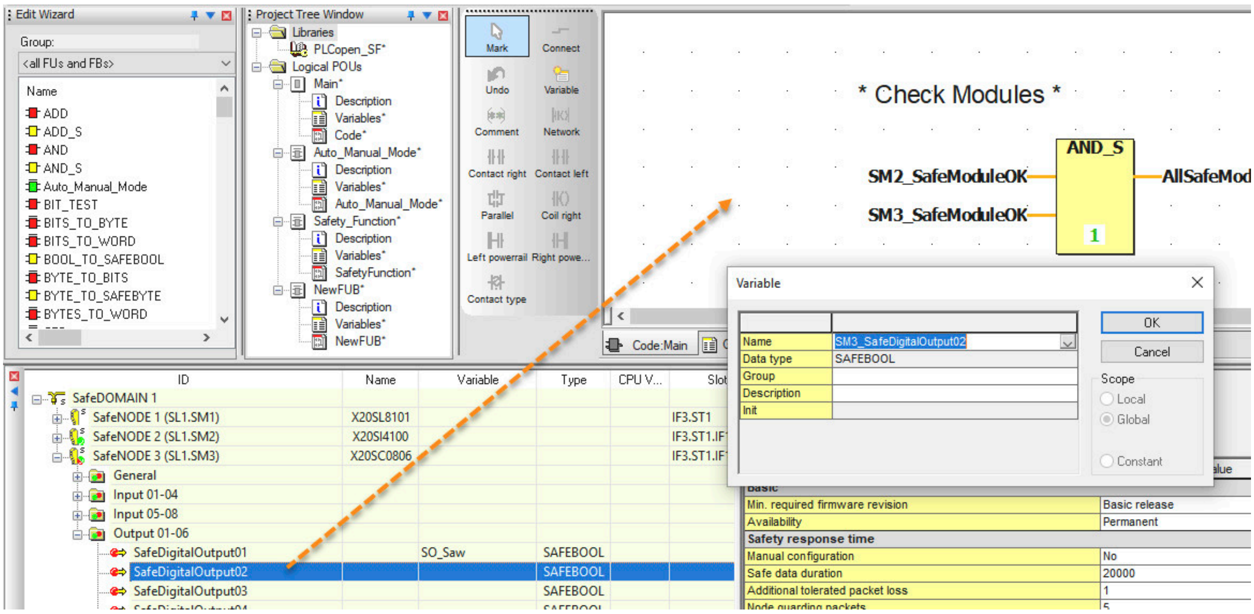
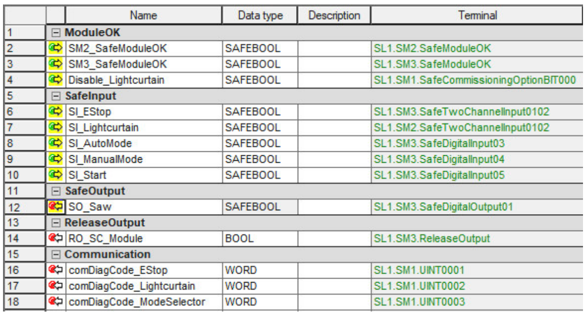


Figure 52: Dragging an I/O channel onto the worksheet - Filling in the declaration dialog box

This button is used to switch to the global variable declaration.

Figure 53: Opening global declarations

Column "Terminal" contains information about which I/O channel the variable is connected to.



	Name	Data type	Description	Terminal
1	ModuleOK			
2	SM2_SafeModuleOK	SAFEBOOL		SL1.SM2.SafeModuleOK
3	SM3_SafeModuleOK	SAFEBOOL		SL1.SM3.SafeModuleOK
4	Disable_Lightcurtain	SAFEBOOL		SL1.SM1.SafeCommissioningOptionBIT000
5	SafeInput			
6	SI_EStop	SAFEBOOL		SL1.SM3.SafeTwoChannelInput0102
7	SI_Lightcurtain	SAFEBOOL		SL1.SM2.SafeTwoChannelInput0102
8	SI_AutoMode	SAFEBOOL		SL1.SM3.SafeDigitalInput03
9	SI_ManualMode	SAFEBOOL		SL1.SM3.SafeDigitalInput04
10	SI_Start	SAFEBOOL		SL1.SM3.SafeDigitalInput05
11	SafeOutput			
12	SO_Saw	SAFEBOOL		SL1.SM3.SafeDigitalOutput01
13	ReleaseOutput			
14	RO_SC_Module	BOOL		SL1.SM3.ReleaseOutput
15	Communication			
16	comDiagCode_EStop	WORD		SL1.SM1.UINT0001
17	comDiagCode_Lightcurtain	WORD		SL1.SM1.UINT0002
18	comDiagCode_ModeSelector	WORD		SL1.SM1.UINT0003

Figure 54: Global declarations

Safety technology \ mapp Safety \ Programming \ SafeDESIGNER \ Developing a project \ Connecting/Disconnecting process data elements and global I/O variables

5.4.4 Linking an I/O channel with an existing variable

To establish a connection between a global variable and an I/O channel, the I/O channel must first be selected in the Safety View and then dragged and dropped onto the variable in the global variable declaration. The reverse is not permitted.

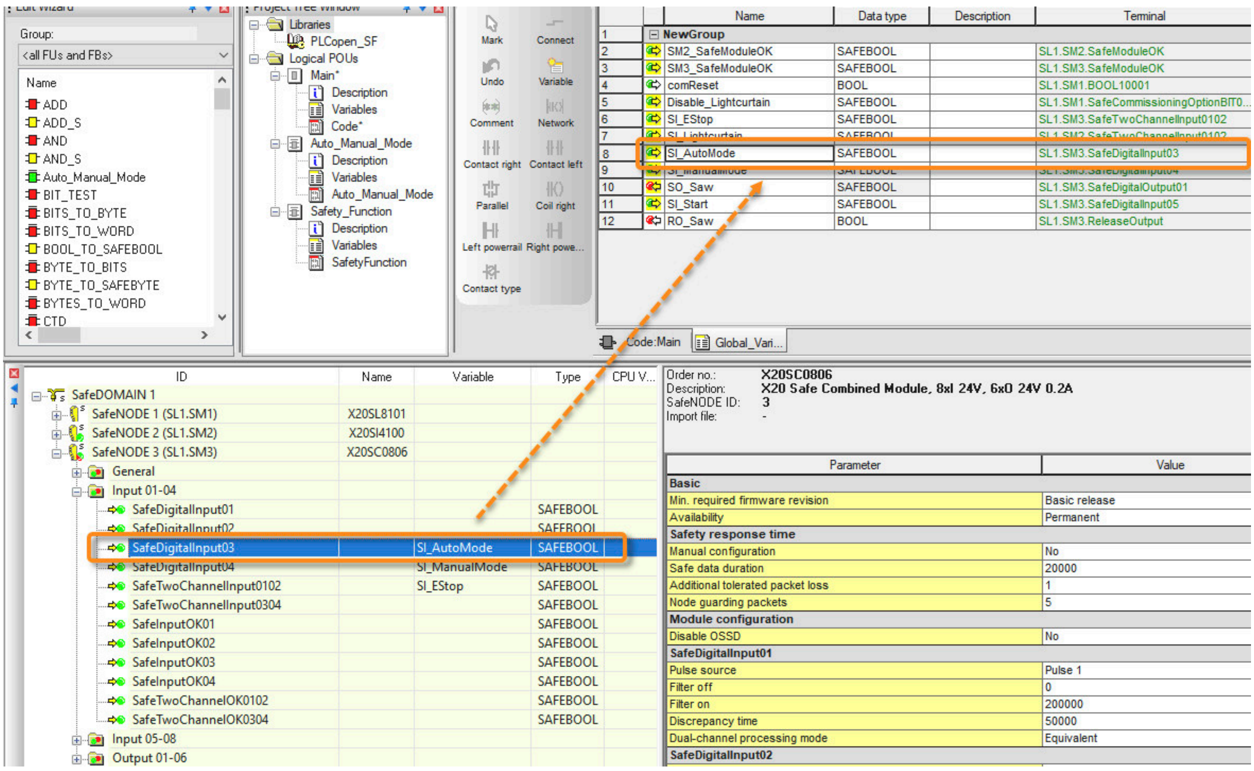


Figure 55: Selecting an I/O channel and assign it to the global variable via drag-and-drop

Safety technology \ mapp Safety \ Programming \ SafeDESIGNER \ Developing a project \ Connecting/Disconnecting process data elements and global I/O variables

Exercise: First safety application

The objective of this exercise is to link two safe inputs to one safe output using an AND operation.

1) Open a SafeDESIGNER project
2) Assign the project password
3) Add function AND_S to the worksheet
4) Declare input "SafeModuleOK" on the SI module as global variable "SM2_SafeModuleOK"
5) Declare input "SafeModuleOK" on the SC module as global variable "SM3_SafeModuleOK"
6) Declaring local variable "AllSafeModulesOK"
7) Connect all variables with an AND_S function

Result:

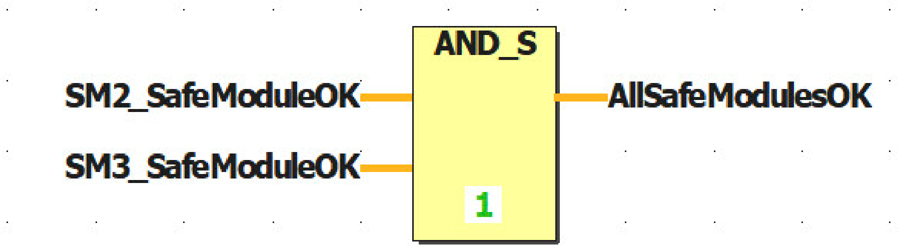


Figure 56: Query the module status

6 Commissioning the safety application

6.1 Compiling a project

The project must be compiled before you can download it. Only projects compiled error-free can be transferred to the SafeLOGIC controller or simulation. The cause of warnings should be identified and corrected.

The "Compile" button is used to compile the safety application and provide it with a unique CRC[5] value.

Figure 57: Compiling

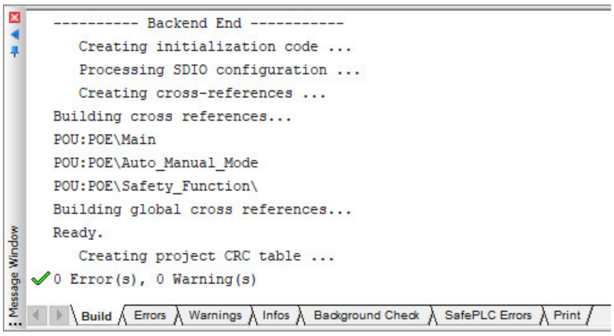


Figure 58: Message window

During the build process, messages, errors and warnings are displayed in the message window. If no errors are reported, the safety application can be transferred to the SafeLOGIC controller.

There are 2 ways to transfer a SafeAPPLICATION:

- Remote Control in SafeDESIGNER (see "Download via the Remote Control dialog box" on page 34)
- mapp Safety HMI application (Download via the mapp Safety HMI application)

Download via "**Remote control**" is a fast method to transfer code changes to the SafeLOGIC controller and test them.
Before commissioning, it is necessary to ensure that a complete and final transfer via the "**mapp Safety HMI**" is carried out.
For the first or final transfer, the "**mapp Safety HMI**" must be selected.

SafeKEY / Safety section

The safety-related data on the SafeLOGIC controller is transferred directly to the SafeKEY and loaded from there.
Since there is no storage medium available on the SafeLOGIC-X controller, a safety section is created in the flash memory of the controller in which the data for the SafeLOGIC-X controller is stored.
This includes the following data on the SafeKEY / Safety section:

- SafeDESIGNER application (application and all SafeDESIGNER parameters for the modules)
- Configuration (unique module ID (UDID), firmware versions of modules)
- SafeOPTION (Safe Commissioning Options, Safe Commissioning Tables, etc.)

Safety technology \ mapp Safety \ Programming \ SafeDESIGNER \ Developing a project \ Compiling a project

5 Cyclic redundancy check (CRC for short) is a procedure for calculating a data checksum

6.2 Download via the Remote Control dialog box

The application can be transferred, messages can be acknowledged and the SafeLOGIC controller can be reset via the Remote Control dialog box.
The Remote Control dialog box is opened in the SafeLOGIC controller shortcut menu in the Safety View.

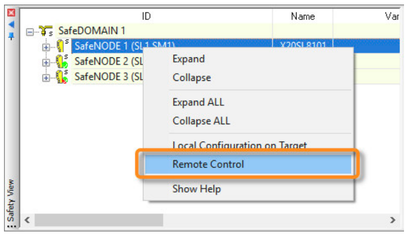


Figure 59: Opening the Remote Control dialog box

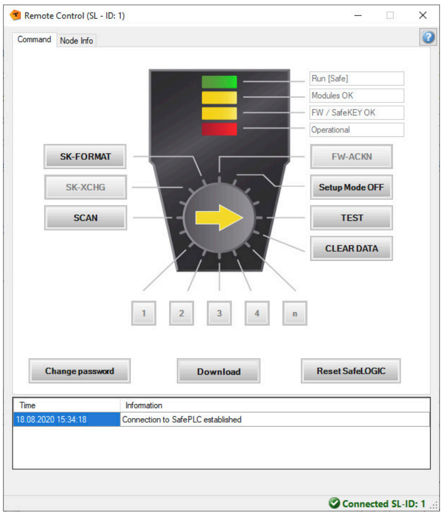


Figure 60: Remote Control dialog box

The SafeKEY password that was defined in the "Config.sfdomain" file in Automation Studio must be entered first.
After the Remote Control dialog box has been opened, the safety application can be transferred to the SafeLOGIC controller via command "Download".
After the download has been completed successfully, the SafeLOGIC controller restarts and the buttons in the Remote Control dialog box appear in orange. This indicates that acknowledgment is required.

For detailed information about the operating and status elements of the Remote Control dialog box, see 9.2 "Operating and status elements" on page 61.

Safety technology \ mapp Safety \ Programming \ SafeDESIGNER \ Dialog boxes for controlling the safety controller \ Remote Control

6.3 Download via the mapp Safety HMI application

After the SafeAPPLICATION has been created and compiled in SafeDESIGNER, it is stored in the mappSafety folder in the Configuration View in Automation Studio and is stored on the controller when the standard application is downloaded in Automation Studio.
The SafeAPPLICATION can now be selected as the standard application in the .sfdomain file. The SafeAPPLICATION is thus automatically preselected in the "Transfer" area of the mapp Safety HMI application.
URL **<IP Address>:81/index.html?visuId=mappSAFETY** can be used to call the mapp Safety HMI application via a browser[6].

6 The latest version of Google Chrome is preferred

Login

To establish a connection with the SafeLOGIC controller, it is necessary to log in to Automation Studio with the user data from the user configuration.

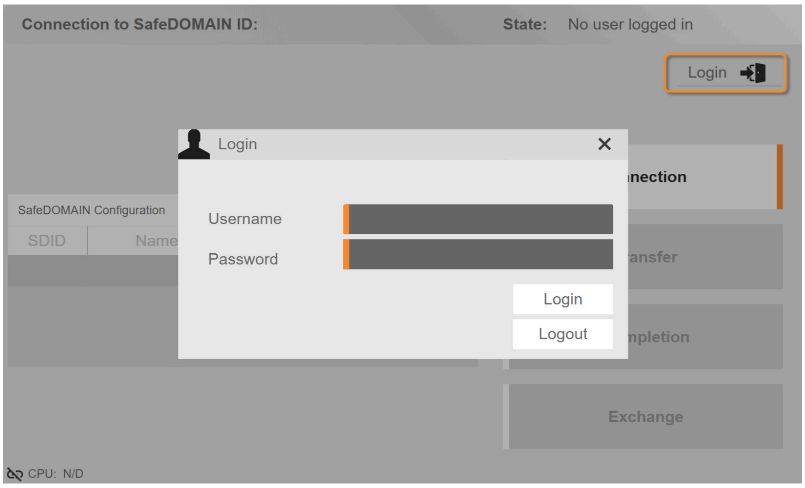


Figure 61: mapp Safety HMI application - Login

Connection

A connection can then be established to the desired SafeDOMAIN by selecting the corresponding entry from the table and confirming via "Connect". When connected, the correct status of the SafeLOGIC controller is displayed.

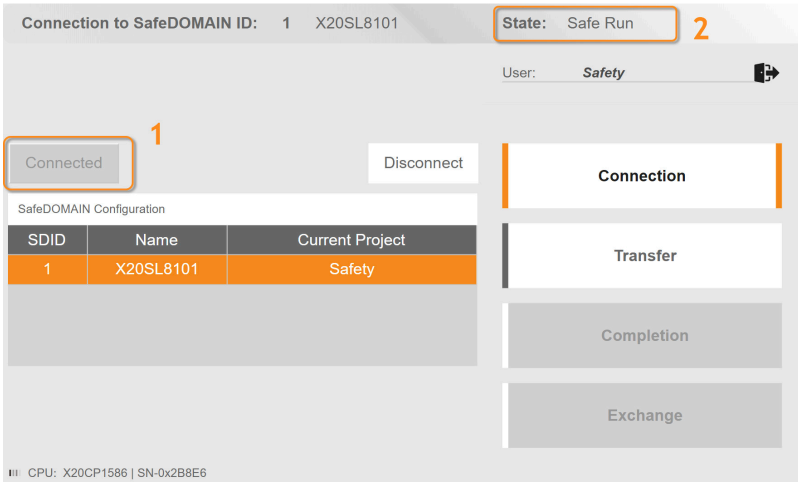


Figure 62: mapp Safety HMI application - Establishing a connection to the SafeLOGIC controller

Downloading the safety application via the mapp Safety HMI application only works if the SafeLOGIC controller is in setup mode. This can be done as follows:

- Enable setup mode in the Remote Control dialog box.
- Empty or formatted SafeKEY (only for SafeLOGIC controller, not for SafeLOGIC-X controller)
- Set the selector switch on the SafeLOGIC controller to an unlabeled position between FW-ACKN and SK-COPY and press and hold the acknowledgment button for approx. 20 seconds (only for SafeLOGIC controller, not for SafeLOGIC-X controller).

Setup mode is only permitted to be enabled during the commissioning of the machine/system. Setup mode must be disabled during operation.

Transfer

The configured standard SafeAPPLICATION is displayed in the "Transfer" area. If it should be transferred, button "Transfer to SafeDOMAIN" is selected directly. Otherwise, another SafeAPPLICATION can be used via "Browse for data".

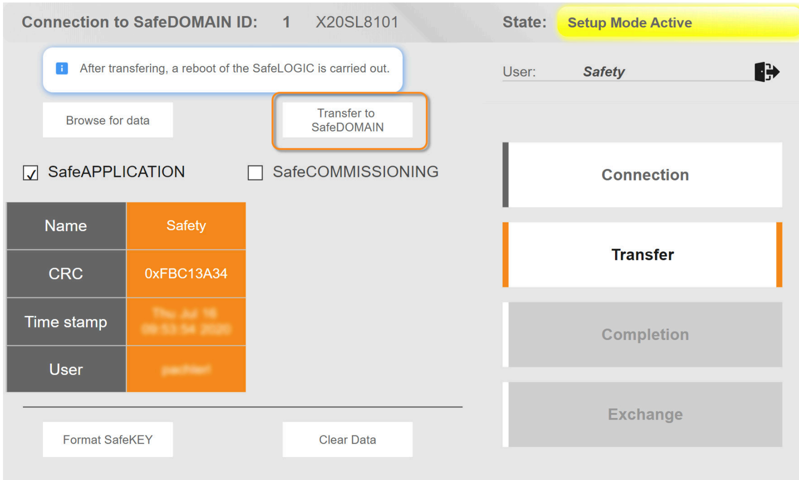


Figure 63: mapp Safety HMI application - Transferring SafeAPPLICATION

Completion

To complete commissioning and exit setup mode, the correctness of the transmitted data must be confirmed. This is done in the "Completion" area. The items to be confirmed are highlighted in white and must be completed one after the other before setup mode can be exited.

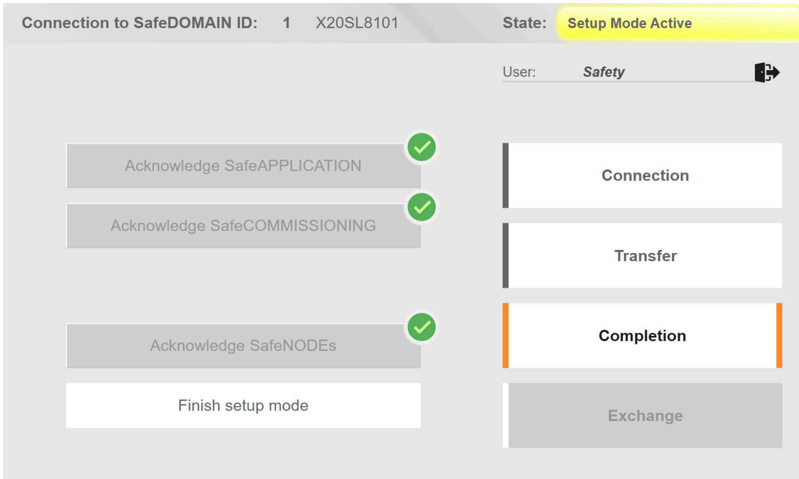


Figure 64: mapp Safety HMI application - Acknowledgment

Afterwards, a complete test of the application is required!

Safety technology \ mapp Safety \ Getting started \ Commissioning the application \ Using the preconfigured HMI application

Exercise: Compiling and downloading

The objective of this exercise is to successfully compile the project and to test the two download options in order to put the SafeLOGIC controller into operation.
Remote Control dialog box

1) Compile the safety application.
2) Opening the Remote Control dialog box
3) Download the safety application to the SafeLOGIC controller.
4) Complete the necessary acknowledgments
5) Check the LED status on the SafeLOGIC controller / the status messages of the Remote Control dialog box.
6) Formats the SafeKEY

mapp Safety HMI

1) Compile the safety application.
2) Compile and download the Automation Studio project.
3) Open the mapp Safety HMI application
4) Logging on and establishing a connection
5) Download the safety application to the SafeLOGIC controller.
6) Complete the necessary acknowledgments
7) Checking the LED status of the SafeLOGIC controller

6.4 Commissioning checklist

The following is a list of the most important steps for commissioning the SafeLOGIC controller / SafeLOGIC-X controller, the SafeIOs and the safety application.

Procedure	Note
Download the Automation Studio project	Download configuration of the modules and mapp Safety When formatting the CPU memory, reset the password for the SLX module.
Wait for startup	**The system performs the following steps:** • Start default CPU • Start SafeLOGIC controller / SafeLOGIC-X controller • Install firmware updates on the modules • Initialize the hardware
Download the program	Transfer via "Download" button in the Remote Control dialog box Safety controller restarts
	In mapp Safety HMI application: • Select SafeAPPLICATION and SafeCOMMISSIONING file • Transfer via button "Transfer to SafeDOMAIN" Safety controller restarts
Acknowledge SafeKEY	LED FW_ACKN lights up permanently Acknowledge via SK_XCHG
Acknowledge new modules	LED MXCHG blinks Acknowledge new modules with 1, 2, 3, 4 or n
Acknowledge firmware	LED FW_ACKN blinks Acknowledge new firmware via FW_ACKN
Test application	Status LEDs light up green Test application

Table 4: Commissioning checklist

Commissioning in setup mode

Setup mode facilitates commissioning since the user can stabilize the machine when setup mode is active, test firmware updates or make changes. In addition, acknowledgment requirements "SafeKEY exchange", "Firmware acknowledge" and "UDID mismatch" are no longer required.
When commissioning is complete, setup mode is exited and a final overall test of the machine is required.
Active setup mode is indicated by both the FAILSAFE LED (X20SL81xx series) or SE LED (X20SLXxxx series) as well as an entry in the logbook.
When setup mode is active, acknowledgment requests "SafeKEY Exchange", "Firmware Acknowledge" and "UDID Mismatch" are no longer necessary.

Setup mode can be enabled and disabled using the operating elements of the "Remote Control" in SafeDESIGNER (X20SL81xx and X20SLXxxx series) or using the selector switch and acknowledgment button (X20SL81xx series). (9.2.2 "Operating elements" on page 63)

Safety technology \ mapp Safety \ Programming \ SafeDESIGNER \ Commissioning the safety controller
Safety technology \ mapp Safety \ Engineering \ SafeLOGIC \ Software functions \ Setup mode

7 Implementing the safety application

7.1 Using PLCopen function blocks

SafeDESIGNER provides various PLCopen function blocks that are used to implement the SafeAPPLICATION. In these function blocks, the input signal is processed, the restart behavior is configured and the reset signal is connected. The function blocks can be added to the worksheet via drag-and-drop using the Edit wizard and a window is automatically opened in which the instance is created.

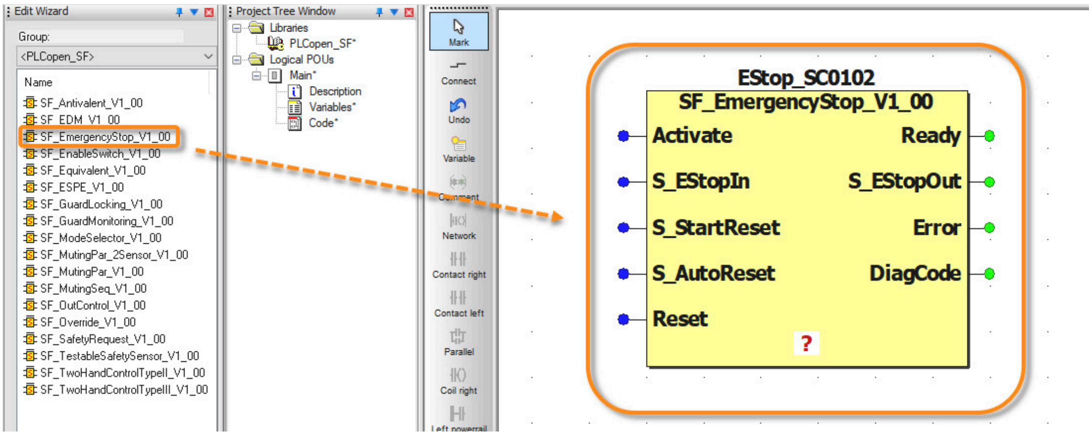


Figure 65: Adding a PLCopen function block for an emergency stop switch

Safety technology \ mapp Safety \ \ Programming \ SafeDESIGNER libraries \ PLCopen_SF \ Function blocks

7.2 Dual-channel evaluation for safe inputs

The safe digital inputs are used for safe sensors and additionally support functions such as dual-channel evaluation and simultaneity monitoring.
The input signals of the signal pairs (e.g. channel 1 and 2) are monitored in the module for simultaneity. The maximum permissible discrepancy time of inputs of a signal pair is configurable.
The safe signal of a dual-channel sensor, such as an emergency stop device or a light curtain, can thus be processed in the safety application via a variable.

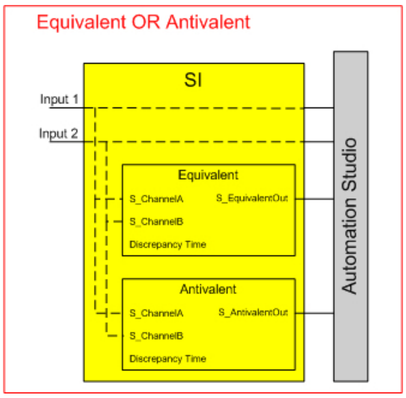


Figure 66: Dual-channel evaluation being performed directly on the safe module

Hardware \ X20 system \ X20 modules \

- Digital input modules \ X20(c)SIx1x0
- Digital mixed modules \ X20SC0xxx

Safety technology \ mapp Safety \ \ Engineering \ SafeIO \ Safe digital inputs

7.3 Pulse source

The module provides pulse signals for diagnosing the sensor line. By default, each pulse signal provides a unique pulse pattern derived from the module's serial number and pulse channel number. This allows any pulse signals to be combined in one signal cable and still cover any cross fault combinations in the cable.

Internal pulse (pulse 1, 2, 3 or 4)

When connecting single-channel or dual-channel electromagnetic switches, each input channel is assigned a dedicated clock output.
It is also possible to select a pulse of another channel in the configuration depending on the wiring.

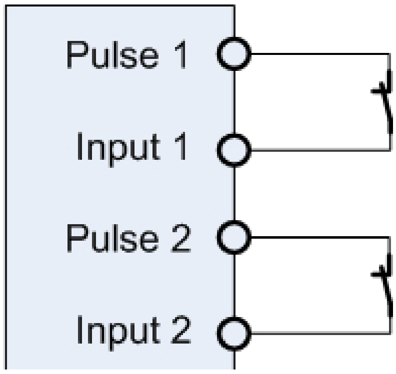


Figure 67: Single-channel connection

No pulse

The pulse check can be disabled to connect electronic sensors with separate line monitoring (OSSD signals).

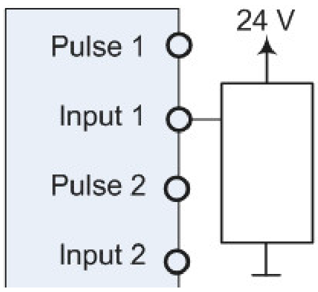


Figure 68: Single-channel connection - Sensors with their own OSSD signal

If a clock pattern from the module is not used, the safe input module itself has reduced error detection. All potential errors in the wiring must be detected through supplementary measures or by the connected devices.

Safety technology \ mapp Safety \ Engineering \ SafeIO \ Safe digital inputs \ Connection examples

7.4 Switching a safe output

To be able to set an output on the safe output module, "start interlock on error" must first be removed via a positive edge of "ReleaseOutput". This must be done once per module after a module or network error.
After that, the output can be switched using the "SafeDigitalOutputxx" outputs in SafeDESIGNER and/or "DigitalOutputxx" in Automation Studio.
If this sequence is not observed, the output channel remains inactive.

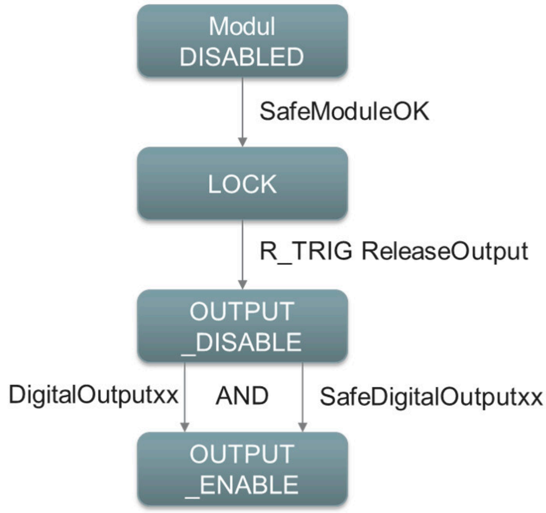


Figure 69: Sequence for switching a safe output

The release signal is a non-safe data type and can also be a signal from a digital input module in Automation Studio, for example.

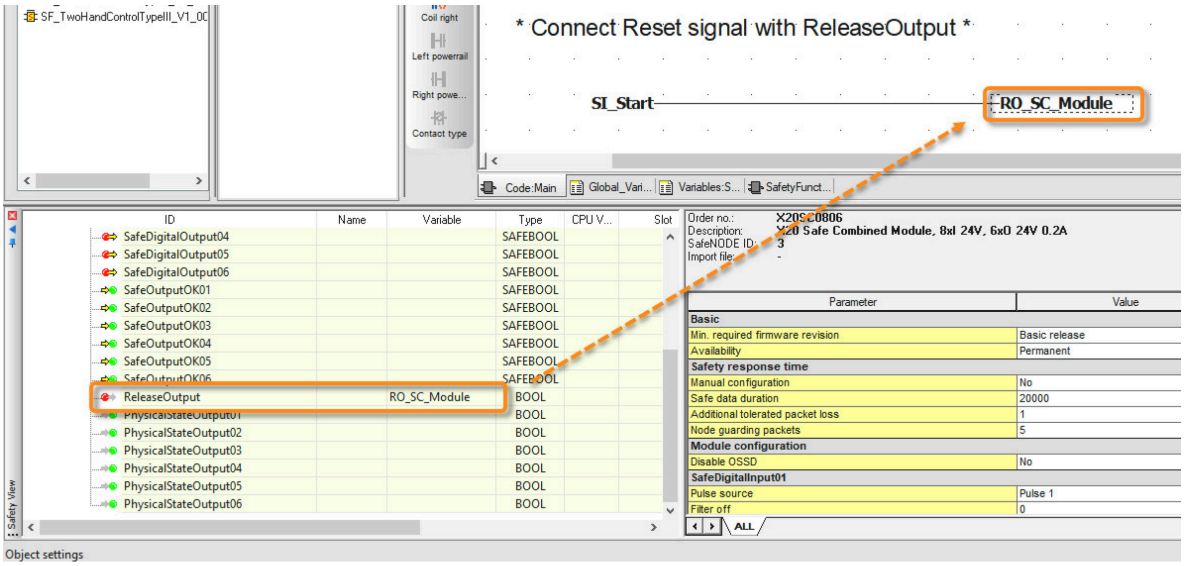


Figure 70: ReleaseOutput

> Safety technology \ mapp Safety \ SafeIO \ Safe digital outputs \ "start interlock on error" state diagram

Exercise: Implement emergency stop function

The objective of this exercise is to use a PLCopen function block. This requires configuration of dual-channel evaluation, switching of a safe output and selection of the correct pulse source for the safe channel.

1) Add function block "SF_EmergencyStop_V1_00"

2) Declare SafeTwoChannelInput on the SC module as global variable "SI_EStop"

3) Configure dual-channel evaluation (equivalent and discrepancy time = 50000 µs)

4) Declare SafeInput on the SC module as global variable "SI_Start"

5) "Change "PulseSource" from "SI_Start" to "Pulse 4"

6) Declare SafeOutput on the SC module as global variable "SO_Saw"

7) Connect function block "SF_EmergencyStop_V1_00"

8) Connect ReleaseOutput on the SC module to "SI_Start"

Result:

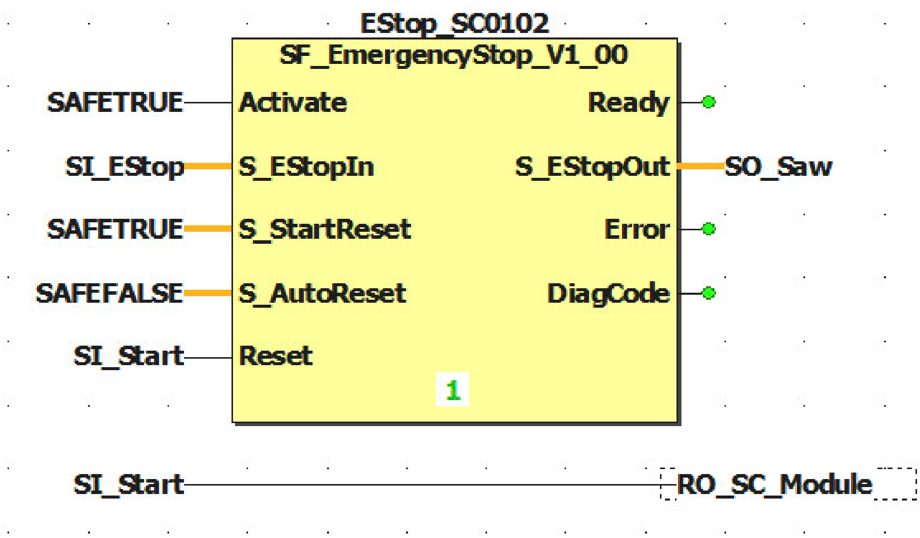


Figure 71: Implementation of the emergency stop function

7.5 Switch-on and switch-off filter

Safe digital inputs are equipped with filters that are individually configurable for switch-on and switch-off behavior. Switch-on filters are used to filter out signal disturbances.
Switch-off filters are used to smooth testing gaps in external signal sources – i.e. OSSD signals – so that unintended cutoffs can be avoided.

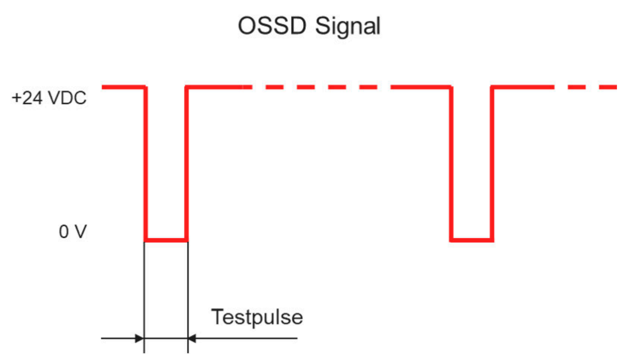


Figure 72: Testing gap

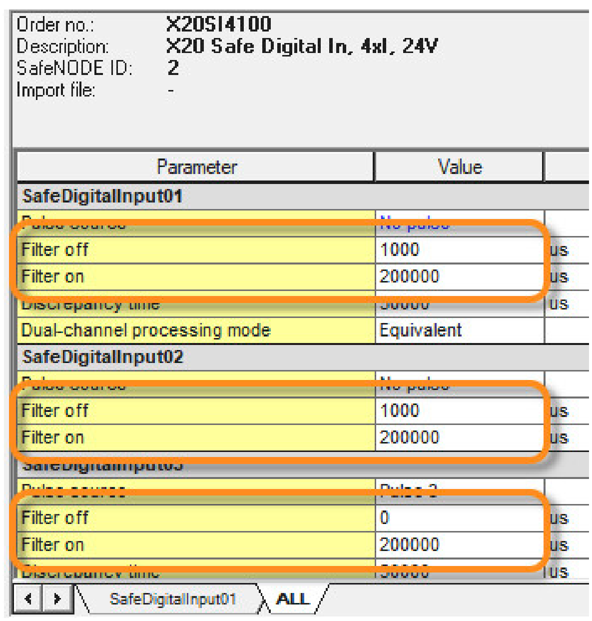


Figure 73: Filter settings for safe inputs

Safety technology \ mapp Safety \ Engineering \ SafeIO \ Safe digital inputs

Exercise: Implement light curtain function

The objective of this exercise is to use a PLCopen function block, configure dual-channel evaluation, make settings for an OSSD signal and link the emergency stop and light curtain signals.

1) Add function block "SF_ESPE_V1_00"

2) Declare SafeTwoChannelInput0102 on the SI module as global variable "SI_Lightcurtain"

3) Configure dual-channel evaluation (equivalent and discrepancy time = 50000 µs)

4) "Change "PulseSource" to "No Pulse"

5) Change the "Filter_off" setting to value 1000 µs

6) Connect function block "SF_ESPE_V1_00"

7) Link the output signal of the emergency stop and light curtain with an AND_S function and connect to "SO_Saw"

Result:

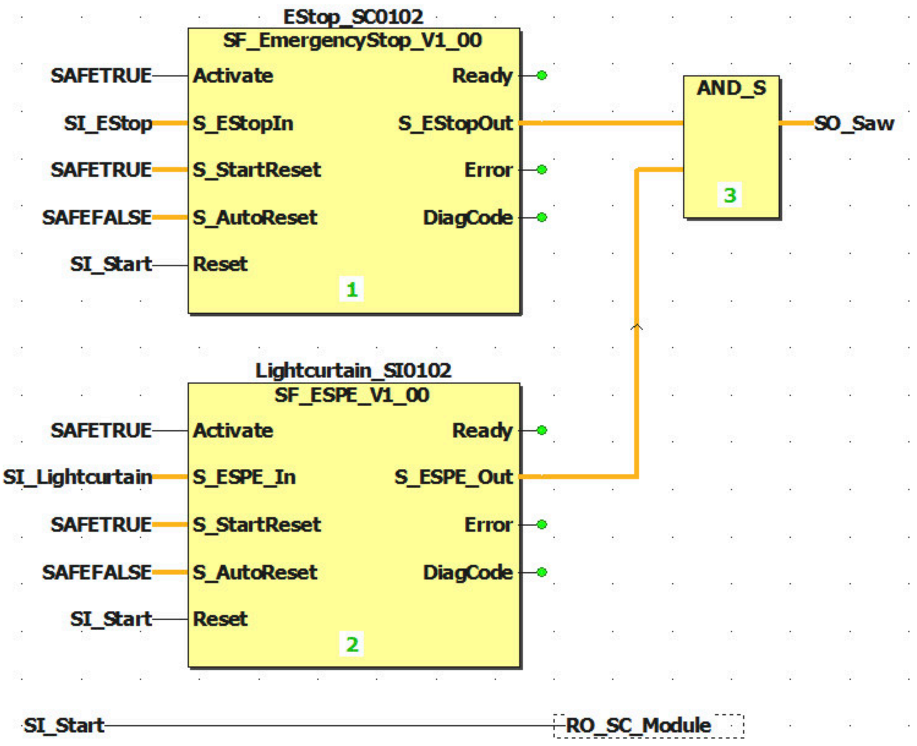


Figure 74: Implementing the light curtain function

7.6 Communication channels between CPU and SafeLOGIC controller

To exchange data between the CPU and the SafeLOGIC controller, communication channels with different data types are available.
The status of the inputs and outputs of the safe modules can be read directly in Automation Studio and evaluated in a program using connected variables. The communication channels are thus only used to exchange internal data of the safety application.

The number of channels is defined in the SafeLOGIC configuration in Automation Studio. The corresponding interfaces are available as inputs and outputs and can be connected to variables in the I/O mapping in Automation Studio and in the Safety View in SafeDESIGNER.

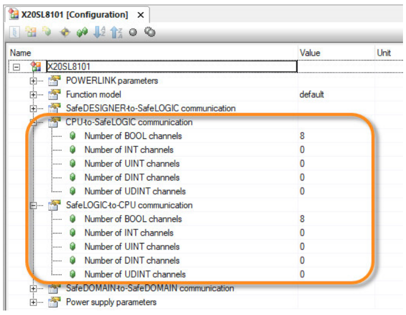


Figure 75: Setting for communication channels in Automation Studio

These interfaces are non-safe data types and are used, for example, to exchange diagnostic codes of PLCopen function blocks or to use a non-safe input for the reset signal.

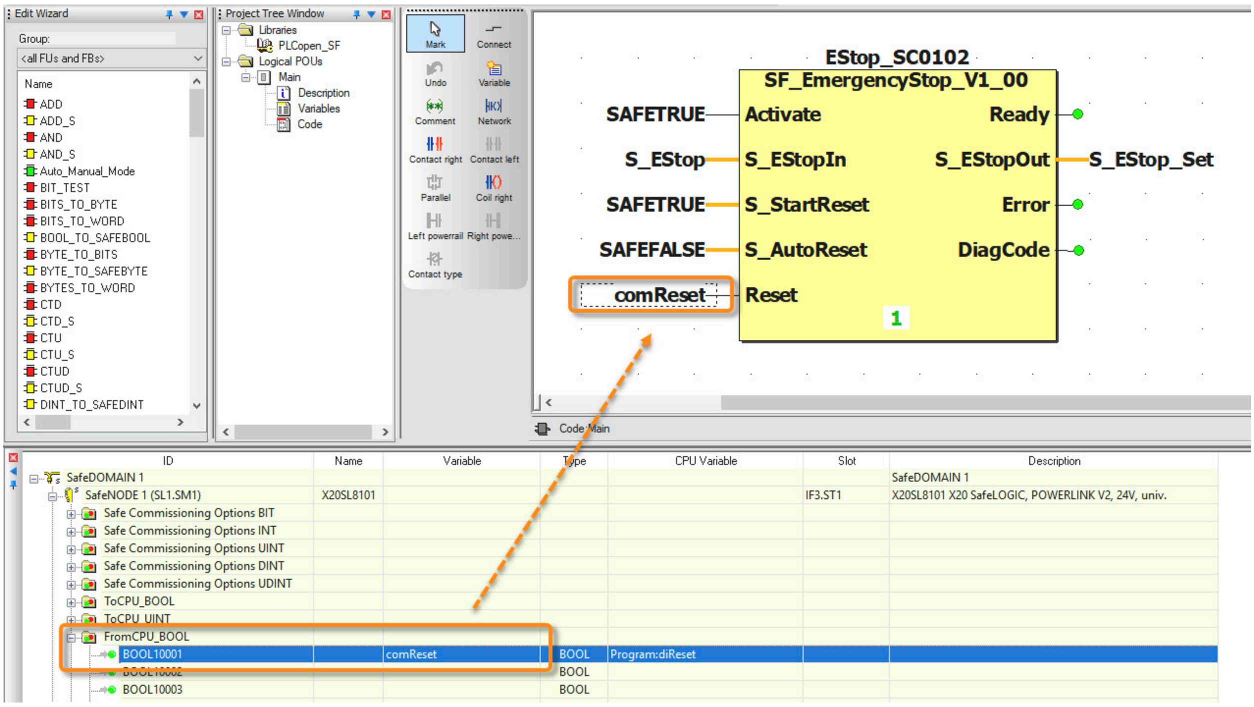


Figure 76: Using communication channels in SafeDESIGNER

Hardware \ X20 System \ X20 modules\ CPUs \ X20(c)SL81xx \ Register description \ Parameters in the I/O configuration

Exercise: Use communication channels and implement mode selector switch

The goal of this exercise is to implement communication channels between the CPU and SafeLOGIC controller to exchange the diagnostic codes and to evaluate the result of these.

Automation Studio

1) Add a new ST program

2) Declare the following variables and call them in the ST program:

Variable name	Data type	Description
diagCode_EStop	UINT	PLCopen diagnostics code Function blocks in SafeDESIGNER
diagCode_Lightcurtain	UINT	

Table 5: Variables in Automation Studio

Variable name	Data type	Description
diagCode_ModeSelector	UINT	
errorInSafetyApplication	BOOL	Result of diagnostic code query

Table 5: Variables in Automation Studio

3) In the SafeLOGIC configuration, enable three "Communication from SafeLOGIC to CPU" channels with data type UINT.

4) Connect variables for diagnostic codes with communication channels in the I/O mapping for the SafeLOGIC controller.

5) Compile and transfer the project.

Result:

```
PROGRAM _CYCLIC

    IF  diagCode_EStop = 32768 AND            //16#8000
        diagCode_Lightcurtain = 32768 AND     //16#8000
        diagCode_ModeSelector = 32768 THEN    //16#8000

        errorInSafetyApplication := FALSE;
    ELSE
        errorInSafetyApplication := TRUE;
    END_IF;

END_PROGRAM
```

Figure 77: Program in Automation Studio

UINT0001	::Program:diagCode_EStop	UINT
UINT0002	::Program:diagCode_Lightcurtain	UINT
UINT0003	::Program:diagCode_ModeSelector	UINT

Figure 78: SafeLOGIC controller I/O mapping

SafeDESIGNER

1) Add function block "SF_ModeSelector_V1_00"

2) Declare SafeDigitalInputs on the SC module as global variable "SI_AutoMode" and "SI_ManualMode"

3) Change PulseSource from "SI_ManualMode" to "Pulse3"

4) Connect function block "SF_ModeSelector_V1_00"

5) Declare "ToCPU_UINT" channels on the SafeLOGIC controller as global variables

6) Connect variables for the diagnostic codes of all function blocks with output "DiagCode"

7) Implement automatic and manual mode

Result:

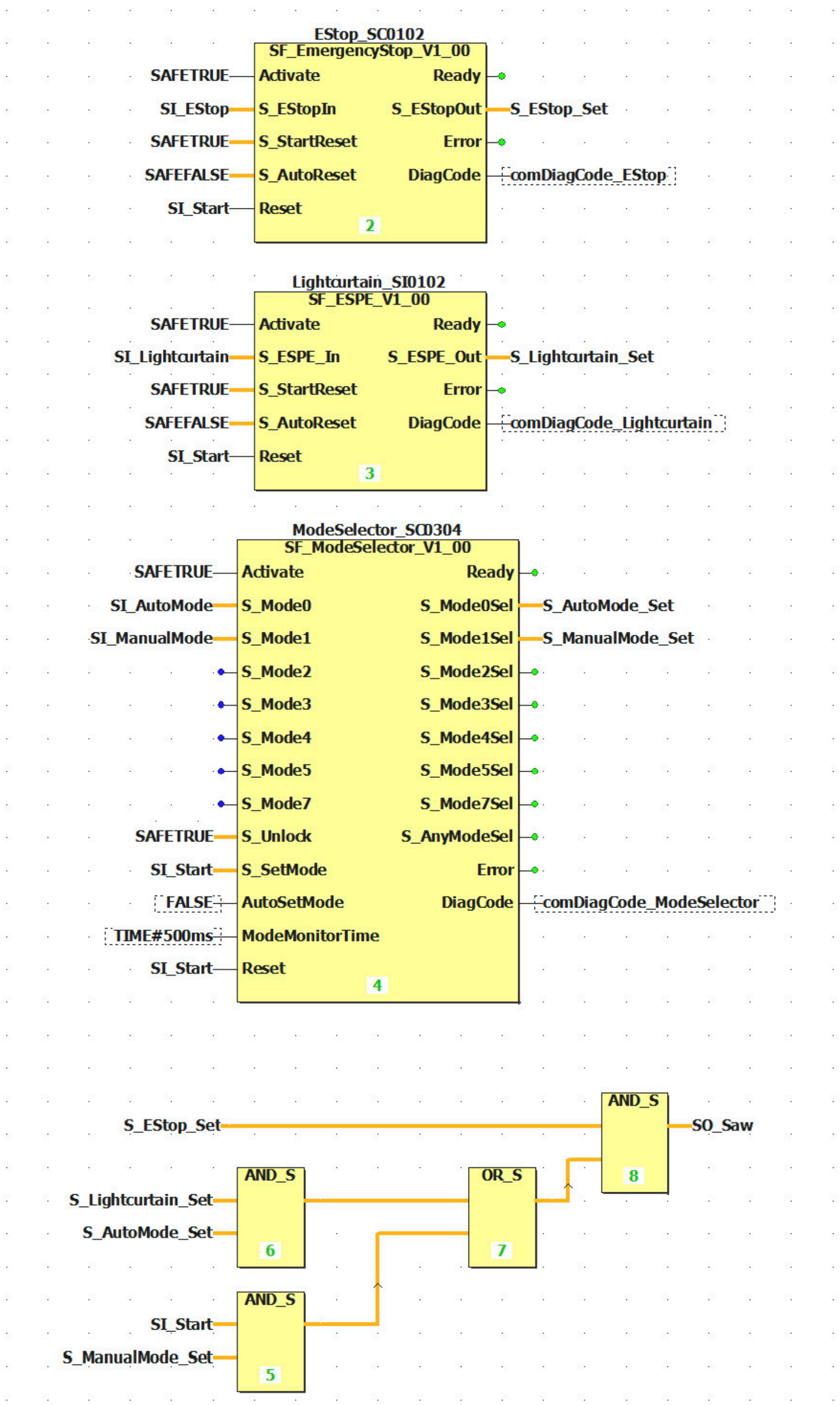


Figure 79: Implementing the mode selector switch function

7.7 Creating a user function block

It is possible to create user function blocks via the shortcut menu of the project tree in SafeDESIGNER.

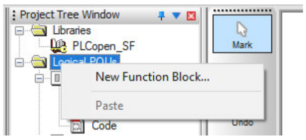


Figure 80: Creating a new function block

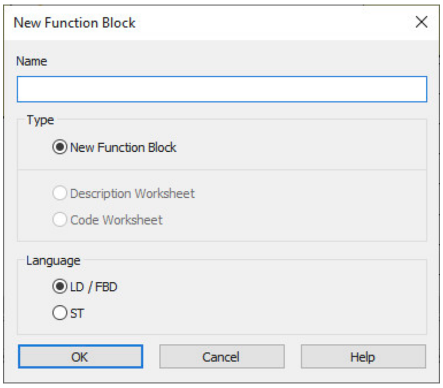


Figure 81: Selecting the programming language for the function block

A dialog box appears in which the name and programming language must be defined. Ladder Diagram, Function Block Diagram and Structured Text are available for selection. After the dialog box has been confirmed, the function block can be opened and edited in the project tree.

A worksheet and a variable declaration are created for each function block.
VAR, VAR_INPUT and VAR_OUTPUT usage can be selected for local variables. Variables used as VAR_INPUT and VAR_OUTPUT represent the inputs and outputs of the function block. The function block is supplied with data via these inputs and outputs.

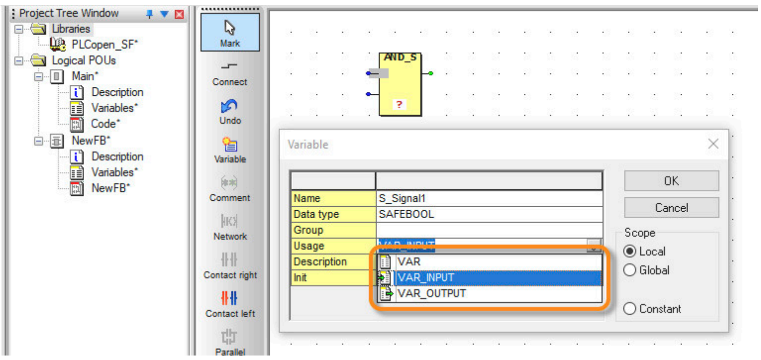


Figure 82: Defining instance variables 'VAR_INPUT, VAR_OUTPUT"

> Each function block is programmed in a separate worksheet. It is possible to directly access I/O variables. I/O variables must be declared globally.

Using a user function block

After the function block code has been compiled, it can be added to the worksheet using the Edit wizard. To facilitate the search function of the Edit wizard, the project name can be selected during grouping.

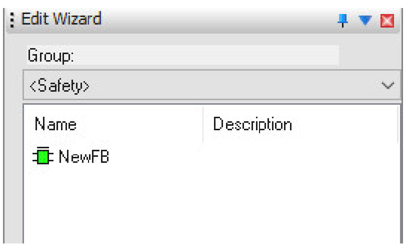


Figure 83: Adding a function block in the Edit wizard via drag-and-drop

Locking a function block

After the functionality of the self-created function block has been tested successfully, the function block should be locked in the shortcut menu via command "Set verification". A CRC value is thus created and the block can no longer be changed.

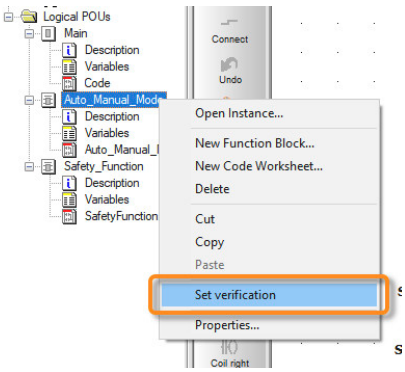


Figure 84: Setting verification

Safety technology \ mapp Safety \ Programming \ SafeDESIGNER \

- User interface \ Project tree - Overview
- Developing a project \ Adding, deleting and renaming function block POUs

Exercise: Create a user function block

The objective of this exercise is to simplify the code in worksheet "Main" and to implement the functions as user function blocks.

1) Implement function block "Safety_Function" for emergency stop, light curtain and mode selector switch

2) Implement function block "Auto_Manual_Mode" for the various operating modes

Parameter	IN/OUT	Description
S_EStop	IN	Equivalent signal from SC module
S_Lightcurtain	IN	Equivalent signal from SI module
S_AutoMode	IN	Signal from SC module
S_ManualMode	IN	Signal from SC module
S_Start	IN	Start signal from SC module
S_EStop_Set	OUT	Status of emergency stop
S_Lightcurtain_Set	OUT	Status of light curtains
S_AutoMode_Set	OUT	Status indicating if automatic mode was selected
S_ManualMode_Set	OUT	Status indicating if manual mode has been selected
DiagCode_EStop	OUT	Diagnostic code for emergency stop block
DiagCode_Lightcurtain	OUT	Diagnostic code for light curtain block
DiagCode_ModeSelector	OUT	Diagnostic code for operating mode block

Table 6: Overview of the function block parameters for "Safety_Function"

Parameter	IN/OUT	Description
S_EStop_Set	IN	Status of emergency stop
S_Lightcurtain_Set	IN	Status of light curtains
S_AutoMode_Set	IN	Status indicating if automatic mode was selected
S_ManualMode_Set	IN	Status indicating if manual mode has been selected
S_Start	IN	CPU communication channel

Table 7: Overview of the function block parameters for "Auto_Manual_Mode"

Parameter	IN/OUT	Description
S_Saw	OUT	Status indicating if all security relevant conditions are fulfilled

Table 7: Overview of the function block parameters for "Auto_Manual_Mode"

Result:

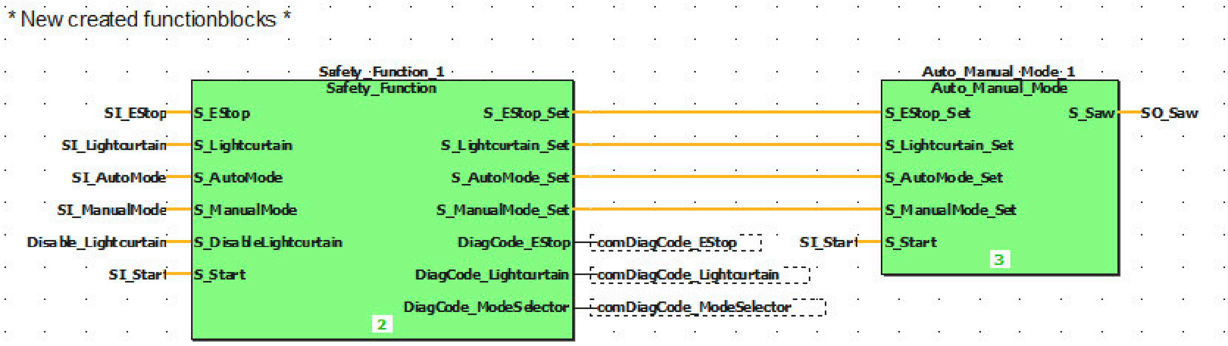


Figure 85: Creating a user function block

8 SafeCOMMISSIONING

SafeCOMMISSIONING makes it possible to adapt the characteristics of the machine at runtime. The following steps are required for this:

Preparing the safety application

For the SafeLOGIC controller, the availability of individual modules can be influenced via parameter "Availability source" and for SafeNODEs via parameter "Availability".
Depending on which SafeCOMMISSIONING file is active, the "Safe Commissioning Options" parameters can be used to change the values of variables according to the configuration.

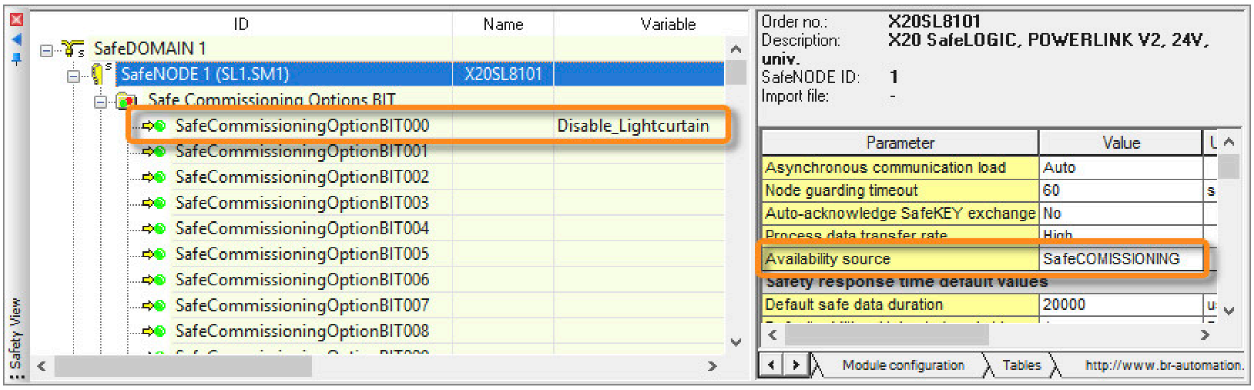


Figure 86: SafeCOMMISSIONING parameters in the "Safety" view

Creating the SafeCOMMISSIONING file

Various SafeCOMMISSIONING files are created and edited via the configurator in the "Project \ SafeCOMMISSIONING" area.

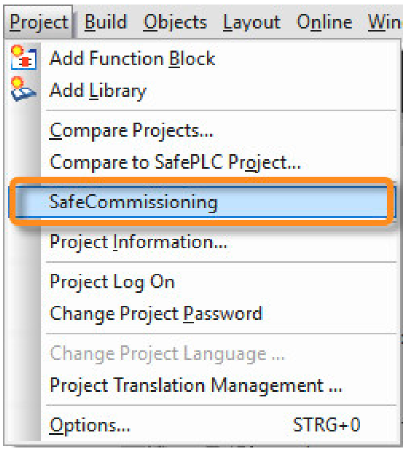


Figure 87: Opening the SafeCOMMISSIONING configurator

Any number of SafeCOMMISSIONING files can be created per application with different configurations. This makes it possible to configure different variants of a machine. In this example, two new SafeCOMMISIONING files are created, one with and one without light curtain evaluation.

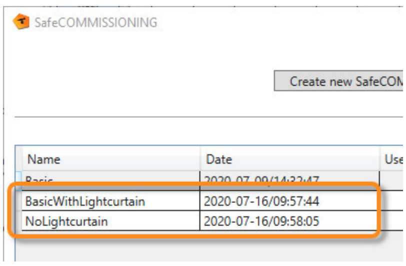


Figure 88: SafeCOMMISSIONING configurator

The parameters of a SafeCOMMISSIONING file can be named and grouped freely. Parameter "SafeOptionBool" is set to either SAFETRUE or SAFEFALSE via property "Enabled". An analog limit value is defined for an integer parameter, for example.

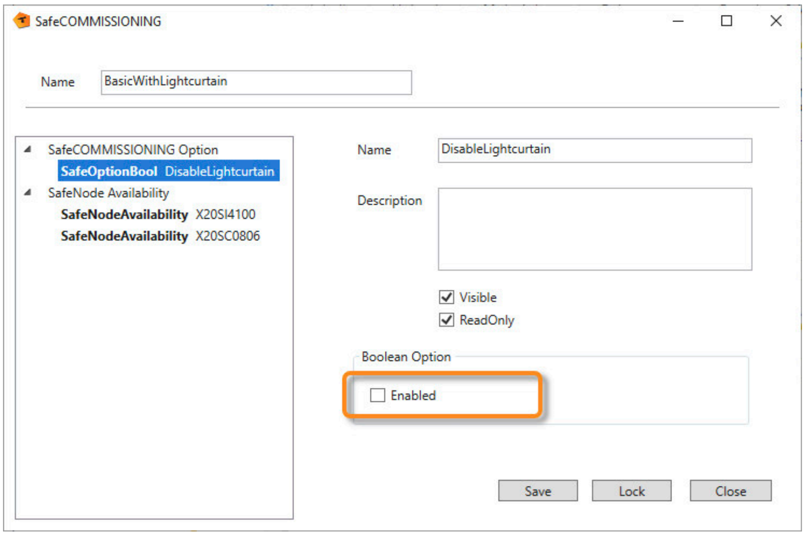


Figure 89: SafeOptionBool

SafeNodeAvailability is used to define for each safe module whether or not the module is available in this machine variant.
The "Lock" button is used to assign a CRC value to the SafeCOMMISSIONING file and create the XML files.

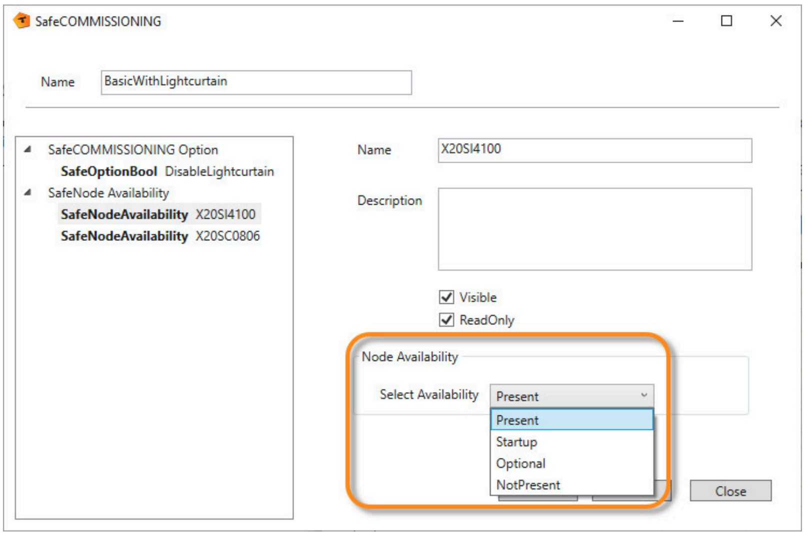


Figure 90: Availability of the SafeNODEs

Configuration in Automation Studio

There are now two .sfopt files in folder "SafeCOMMISSIONING" in the Configuration View. A default SafeCOMMISSIONING file can be selected in the .sfdomin configuration. These changes are then transferred to the controller.

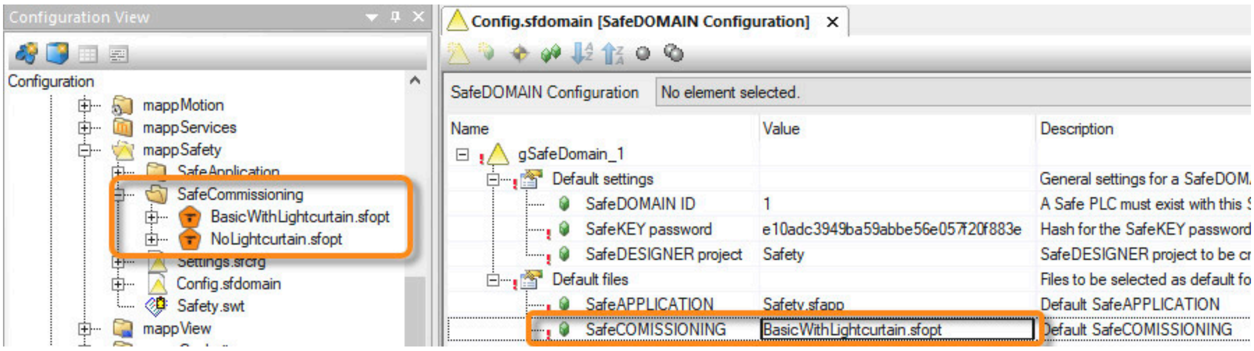


Figure 91: Configuration in Automation Studio

Selection and transfer

The SafeCOMMISSIONING.xml file is available for download in the "Transfer" area of the mapp Safety HMI application. Alternatively, a different SafeCOMMISSIONING.xml file can be selected from the transferred files.

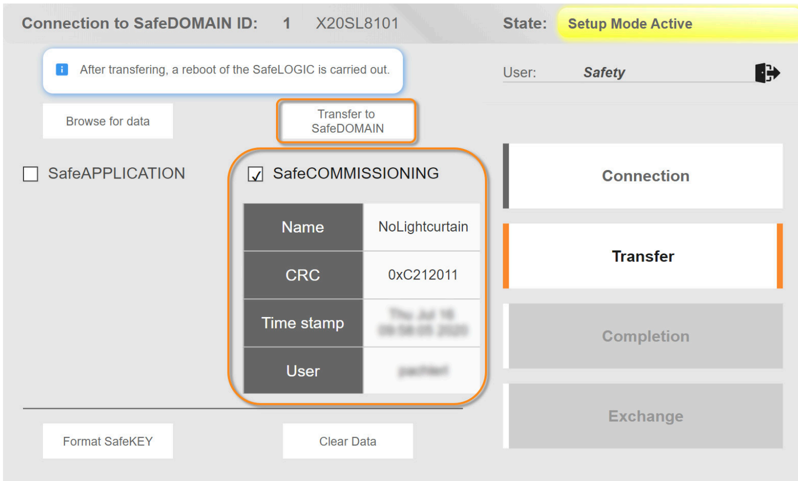


Figure 92: SafeCOMMISSIONING file being transferred via the mapp Safety HMI application

Safety technology \ mapp Safety \ Getting started \ Creating a SafeAPPLICATION \ SafeCOMMISSIONING

Exercise: Implement SafeCOMMISSIONING

The objective of this exercise is to define two different hardware expansion stages and to select these via the mapp Safety HMI application (3.1.2 "Light curtain function" on page 12).

1) Declare SafeCommissioningOptionBIT on the SafeLOGIC controller as "Disable_Lightcurtain"

2) SafeCOMMISSIONING configuration

   BasicWithLightcurtain

   - SafeOptionBool → Disable
   - SafeNodeAvailability SL1.SM2 → Present
   - SafeNodeAvailability SL1.SM3 → Present

   NoLightcurtain

   - SafeOptionBool → Enabled
   - SafeNodeAvailability SL1.SM2 → NotPresent
   - SafeNodeAvailability SL1.SM3 → Present

3) Function block "SafetyFunction" - Create new input variable "S_DisableLightcurtain" and adapt code

4) Download changes to the SafeLOGIC controller via the Remote Control dialog box

5) Change default SafeCOMMISSIONING file in the .sfdomain configuration in Automation Studio

6) Download the Automation Studio project.

7) Open the Remote Control dialog box in SafeDESIGNER and enable setup mode

8) Open the mapp Safety HMI application (http://**IP address**:81/index.html?visuId=mappSAFETY)

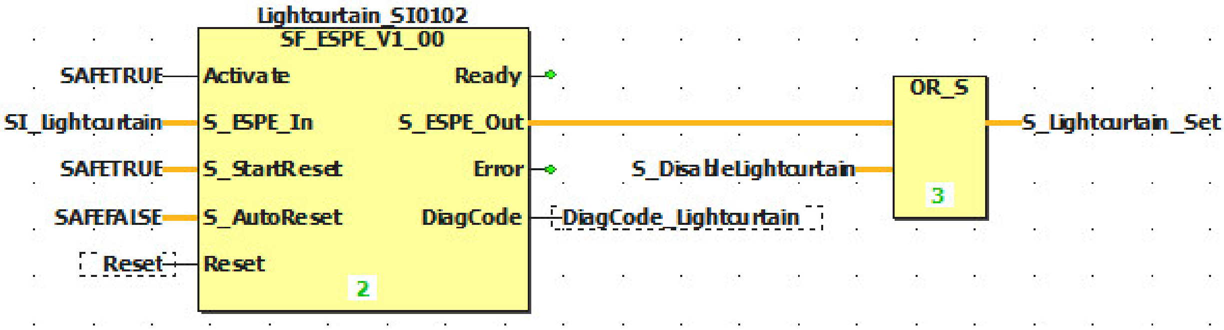
9) Connect to SafeLOGIC controller

10) Click the "Browse for data" button, select the SafeCOMMISSIONING file and transfer it to the SafeDOMAIN

11) Acknowledge SafeCOMMISSIONING and SafeNODEs and exit setup mode

Now you can choose between two SafeCOMMISSIONING files via the mapp Safety HMI application and, depending on which file is active, the function of the light curtain is enabled or disabled.

Result:


Lightcurtain_SI0102
SF_ESPE_V1_00
SAFETRUE
Activate
Ready
SI_Lightcurtain
S_ESPE_In
S_ESPE_Out
SAFETRUE
S_StartReset
Error
SAFEFALSE
S_AutoReset
DiagCode
DiagCode_Lightcurtain
Reset
Reset
2
OR_S
S_DisableLightcurtain
3
S_Lightcurtain_Set

9 Diagnostics and service for safety applications

9.1 Diagnostics for the SafeAPPLICATION

9.1.1 Checking I/O status bits

There is one bit of information about the state of the channel or multi-channel evaluation for each I/O channel. Safe output modules provide additional information about the physical switching state.

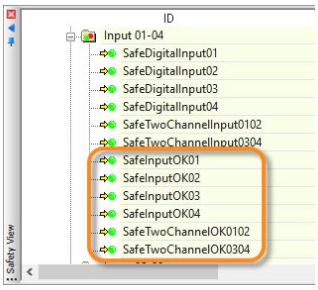


Figure 93: Status information for a safe input module

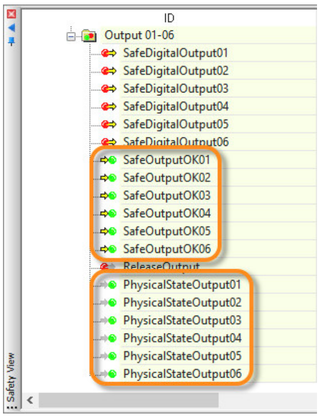


Figure 94: Status information for a safe output module

To receive the status information, the status information channels in the worksheet must be used.

Automation Studio's I/O mapping also provides status information about the safe modules for diagnostics in the form of non-safe data points.

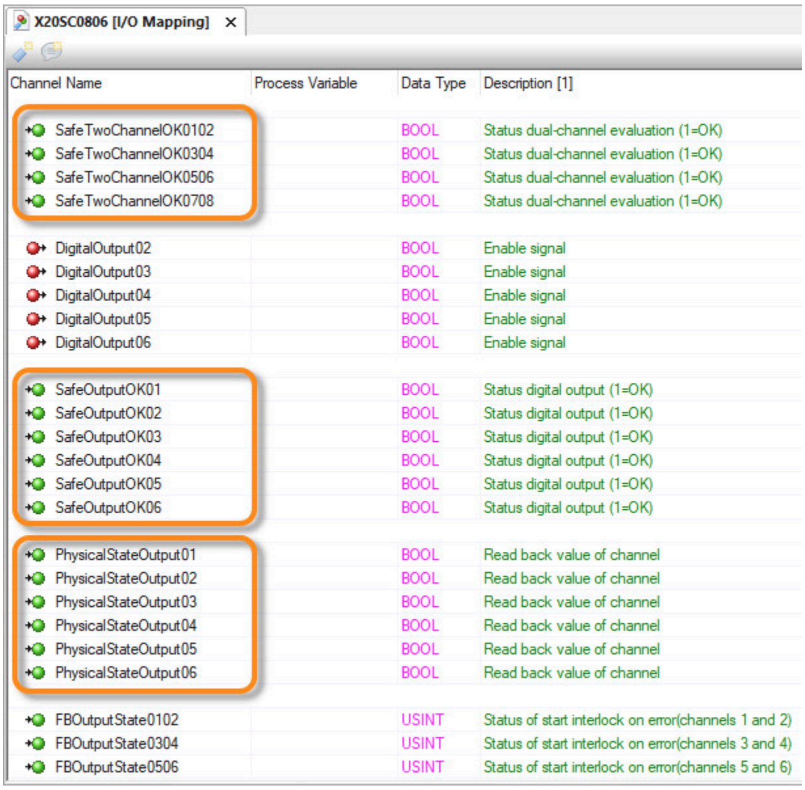


Figure 95: I/O status bits in Automation Studio

9.1.2 Checking the variable status

SafeDESIGNER offers a function to make current values of the variables visible. This function is enabled via the following button:

Figure 96: Button "Variable status"

Variable status in the variable declaration

Button "Global declaration" or "Toggle WS" can be used to switch to the variable declaration. Button "Variable status" is used to display the current values of the variables in column "Online value".

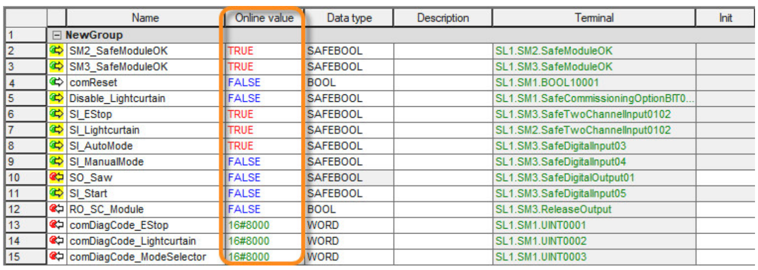


Figure 97: Variable status in the global declaration window

Variable status in the visual editor

The visual editor displays the current value for each variable. The connecting lines between the function blocks are color-coded according to the variable value.

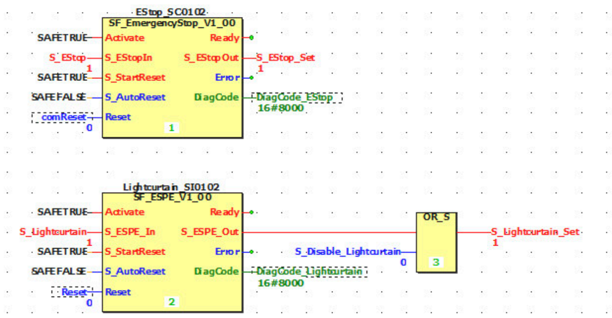


Figure 98: Variable status in the worksheet

Color	Meaning
Red	Signal connected through
Blue	Signal not connected through
Green	Constants and information

Table 8: Color legend in the variable status

Variable status in the Watch window

Each variable can be added to the Watch window via the shortcut menu when the variable status is enabled. In debug mode, variables can be forced in the Watch window. (9.1.3 "Forcing variables" on page 56)

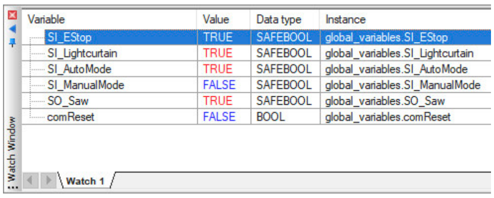


Figure 99: Variable status in the Watch window

Safety technology \ mapp Safety \ Programming \ SafeDESIGNER \ Commissioning the safety controller \

- Monitoring: Displaying the variable status
- Monitoring: Using the Watch window

9.1.3 Forcing variables

To force variable values, debug mode must be enabled on the SafeLOGIC controller. First, the SafePLC dialog box is opened (SafeAPPLICATION password required) and then, debug mode is switched on via button "Debug".

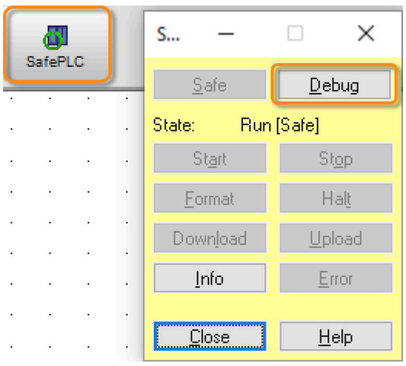


Figure 100: SafePLC dialog box

Forced variables can result in dangerous situations on the machine. Therefore, it must always be ensured that the machine is secured appropriately.

Double-clicking on the variable opens a wizard for enabling the force operation. The action must be acknowledged by the user, whereby an information window draws attention to the possible dangers.

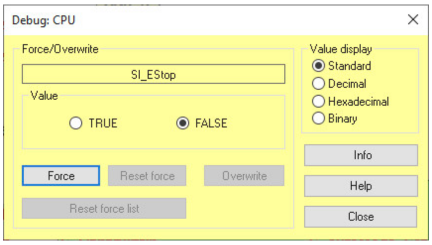


Figure 101: Forcing the variable

In the visual editor, the forced variables are distinguished by a pink background color.

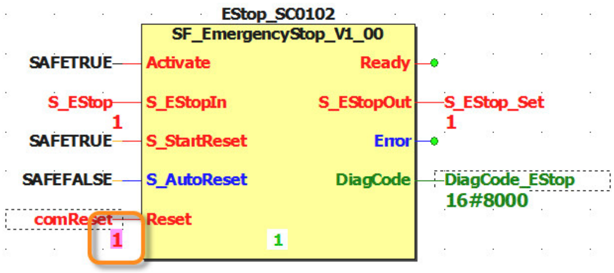


Figure 102: Forced variables

All forced variables are reset when switching to safe mode via the communication window and a message window appears, which must be confirmed.

Safety technology \ mapp Safety \ Programming \ SafeDESIGNER \ Commissioning the safety controller \ Debugging: Forcing, overwriting, single-cycle mode

9.1.4 Cyclic execution of the safety application

SafeDESIGNER allows the safety application to be executed in single cycles for testing purposes. To do so, it is necessary to first switch to debug mode and then to the stop state by clicking on the "Stop" button.
Clicking on the "Single cycle" button will execute the safety application one time.

Figure 103: Stop state

To return to safe mode, first select the "Next" button and then "Safe".

Safety technology \ mapp Safety \ Programming \ SafeDESIGNER \ Commissioning the safety controller

- Debugging: Forcing, overwriting, single-cycle mode
- Dialog boxes for setting the SafePLC \ Dialog box "Debug"

9.1.5 Simulating the application

Simulation can be used to test the safety application independently of the hardware. This virtual SafeLOGIC controller is started via the "Simulate" button.

Figure 104: Starting the simulation

A compilation procedure starts automatically and the simulated SafeLOGIC controller is started. An icon in the "Information" area of the Windows taskbar indicates that the simulation is running.

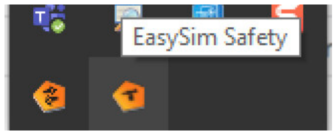


Figure 105: Symbol in system tray

If the simulation is active, the download procedure can only be started via the SafePLC dialog box. For this, it is necessary to first switch to debug mode, then stop the simulation using the "Stop" button and finally start the download. The safety application can now be tested.

Safety technology \ mapp Safety \ Programming \ SafeDESIGNER \ EasySim controller simulation

Exercise: Observe and manipulate application in a simulation and on real hardware

The objective of this exercise is to transfer the application to the simulation, enable the variable status and force values. Afterwards, the simulation is disabled again and the program is observed and manipulated on the real hardware using the variable status.
Diagnostics in simulation

1) Enable simulation (project is compiled automatically)
2) Open "SafePLC" dialog box
3) Select debug mode
4) Stop the SafeLOGIC controller
5) Download project
6) Enable variable status
7) Force values
8) Disable simulation

Diagnostics on real hardware

1) Enable variable status
2) Open "SafePLC" dialog box
3) Select debug mode
4) Force values

9.1.6 Automation Studio Logger window

The system keeps a logbook for safety-related components and for safe communication. The Logger data can be accessed in Automation Studio via main menu "Open \ Logger".

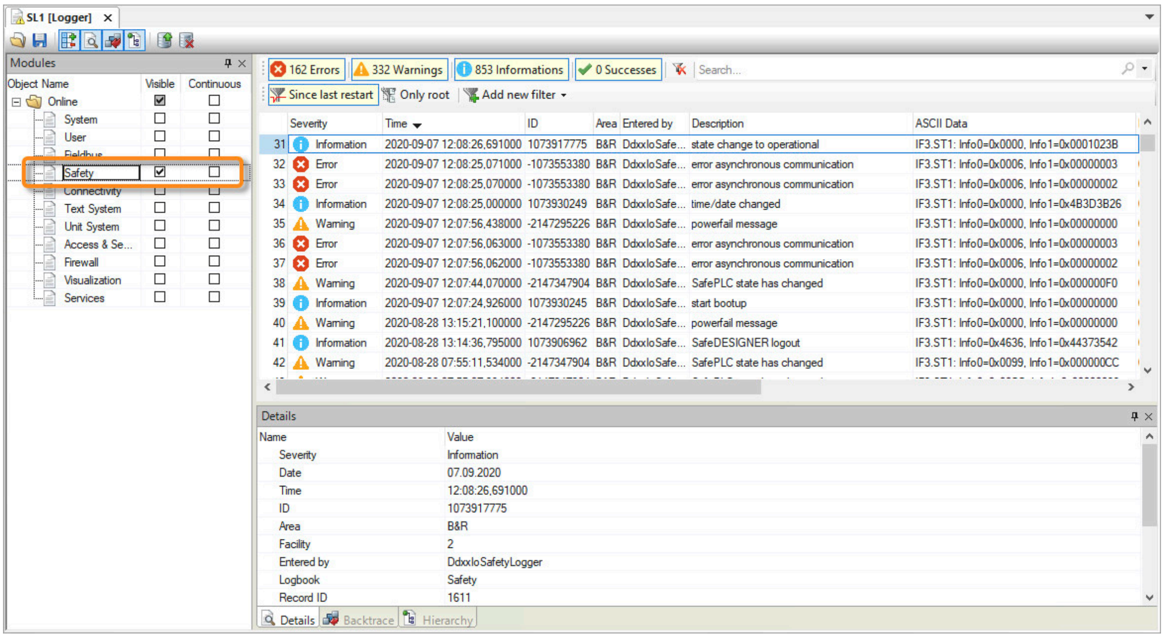


Figure 106: Safety entries in the Automation Studio Logger

The following entries are recorded in this logbook:

- Replacing safety modules
- Configuring safety modules
- Downloading the application
- Updating the firmware
- Changing to the FAIL SAFE state
- Warnings for channel errors
- ...

Diagnostics and service \ Diagnostics tools \ Logger

9.1.7 Project comparison

SafeDESIGNER provides the user with the opportunity to compare projects to each other. This means that the safety application, configuration and parameters of the selected projects are compared.
Two different methods can be selected for project comparison:

- Compare projects...
- Compare to project on the safety controller...

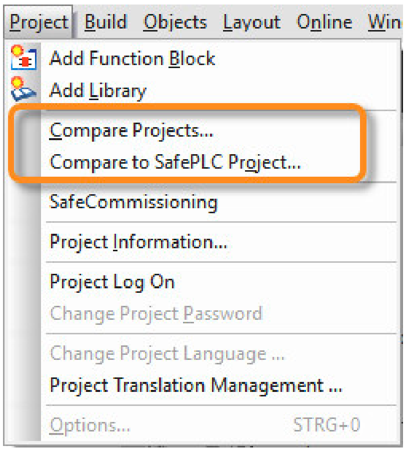


Figure 107: Starting project comparison

Compare projects...

This option is used to compare the project currently opened in SafeDESIGNER with a locally stored project. Window "Project Comparer" opens in which the project to be compared is defined. A file with the extension .swt must be selected.

Compare to project on the safety controller...

The second option is to compare it with a project from the safety controller. To do so, it must be connected and the project sources must have been stored. Under "Online \ Communication parameters" in the lower part of the window, there is an entry that must be selected to download the code to the SafeLOGIC controller. The entry must be deselected if storing is not desired.

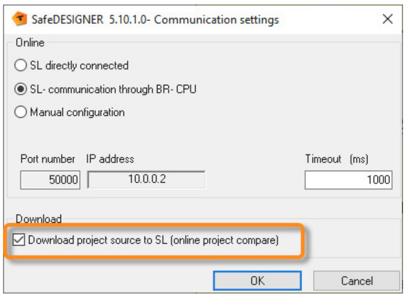


Figure 108: Storing project sources on the safety controller

Project comparison is divided into two areas. In the lower area, the differences are displayed in a tree structure. Selecting an entry opens the corresponding file in the upper area. The differences are shown here and labeled in different colors.

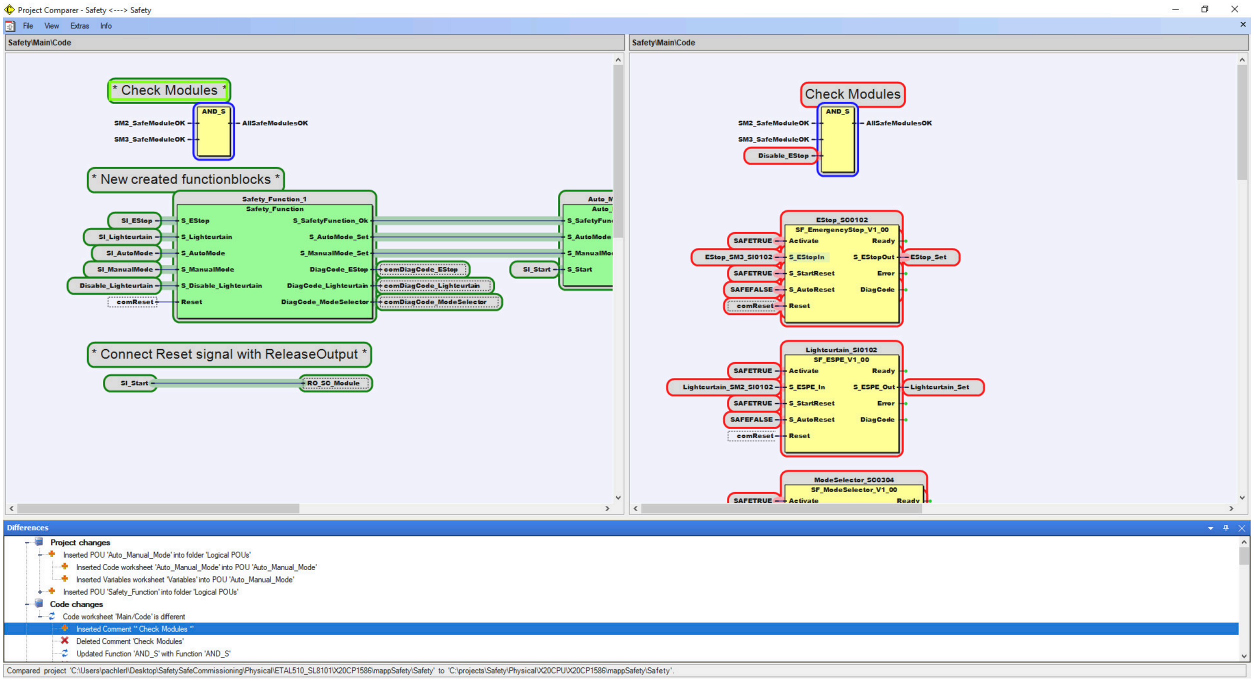


Figure 109: Project comparison in SafeDESIGNER with the current project on the left, the project to be compared on the right

Safety technology \ mapp Safety \ Programming \ SafeDESIGNER \ Introduction \ Comparing projects

9.2 Operating and status elements

SafeLOGIC

The SafeLOGIC controller is equipped with an application interface that contains LED status indicators, a selector switch and a confirmation button. The user can view states via this application interface and perform various actions.

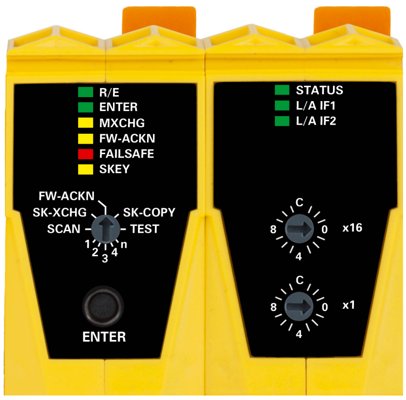


Figure 110: Operating and status elements on the SafeLOGIC controller

Remote Control dialog box

The Remote Control dialog box is an interface that can be used to operate a SafeLOGIC controller and SafeLOGIC-X controller. The current status of the safety controller and various buttons are displayed. The buttons are locked, highlighted or neutral depending on the operating possibility. This prevents incorrect entry.

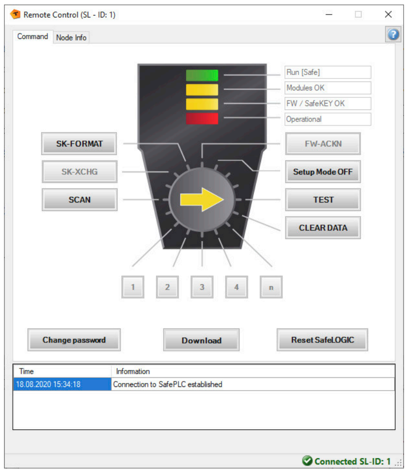


Figure 111: Remote Control dialog box

9.2.1 Status elements

LED status indicators of the SafeLOGIC controller

The LED status indicators of the safety processor are used to indicate the status of the SafeLOGIC controller, safety application and safe modules.
The LED status indicators indicate different operating and error states via different blinking states or continuous illumination.
For a detailed breakdown, see the manual for the SafeLOGIC controller and Automation Help.

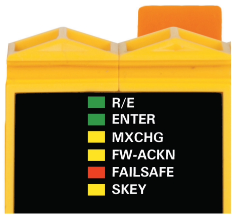


Figure 112: SafeLOGIC

Hardware \ X20 system \ X20 modules \ CPUs \ X20(c)SL81xx \ Operating and connection elements \ Safety processor \ LED status indicators of the safety processor

Status element of the Remote Control dialog box

The status of the SafeLOGIC controller / SafeLOGIC-X controller is displayed in the corresponding text fields in the Remote Control dialog box.

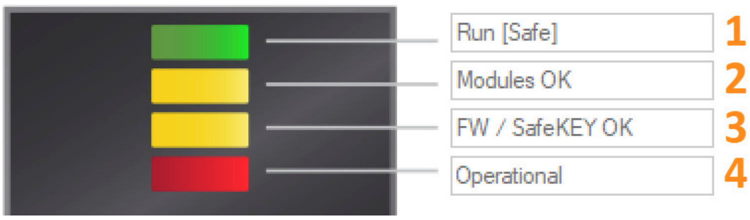


Figure 113: SafeLOGIC

1) SafeOS status → Status of the operating system of the SafeLOGIC controller

2) Module status → Message about the state of the module configuration ("OK", "Acknowledgment required" or "Modules missing")

3) System status → Status of the system ("Firmware / SafeKEY OK" or "Acknowledgment required")

4) Operation mode → Mode of the SafeLOGIC controller ("Operational", "Pre-operational" or "FailSafe")

Safety technology \ mapp Safety \ Programming \ SafeDESIGNER \ Dialog boxes for controlling the safety controller \ Remote Control \ Status indicators for Remote Control

9.2.2 Operating elements

Actions can either be performed directly on the SafeLOGIC controller or via the Remote Control dialog box in SafeDESIGNER.

Entry via SafeLOGIC controller

Confirmation key ENTER on the SafeLOGIC controller must be pressed between 500 milliseconds and 4 seconds later.
This does not apply when the setup mode is enabled or disabled and when the SafeKEY is formatted. In these cases, the button must be pressed for 20 to 30 seconds.
If the entry is correct, the ENTER LED blinks red, whereas blinking tree times indicates an incorrect entry.

Figure 114: Operating elements on the SafeLOGIC controller

Hardware \ X20 system \ X20 modules \ X20 CPUs \ X20(c)SL81xx \ Operating and connection elements \ Safety processor \ Selector switch and confirmation button

Entry via Remote Control dialog box

Using the Remote Control dialog box prevents a wrong entry. All buttons that are not permitted to be selected at the respective time are disabled.

Safety technology \ mapp Safety \ Programming \ SafeDESIGNER \ Commissioning the safety controller \ Dialog boxes for controlling the safety controller \ Remote Control

Description of operating elements

SafeLOGIC	Remote Control	Description
FW-ACKN	FW-ACKN	Confirms the firmware
Between FW-ACKN and SK-COPY[7]	Setup Mode on/off	Enables and disables the setup mode[8]
TEST	TEST	Performs an LED test
-	CLEAR DATA	Deletes subsequently loadable data from the SafeKEY
1, 2, 3, 4, n	1, 2, 3, 4, n	Acknowledges new modules
SCAN	SCAN	Triggers a module scan
SK-XCHG	SK-XCHG	Acknowledges the SafeKEY
Between SK-XCHG and FW-ACKN[1]	SK-FORMAT	Formats the SafeKEY

Table 9: Operating elements for the SafeLOGIC controller and Remote Control dialog box

7 The acknowledgment button must be pressed for 20 to 30 seconds.

8 The setup mode must be disabled again after commissioning has been completed.

SafeLOGIC	Remote Control	Description
-	Change password	Changes the SafeKEY password
-	Download	Downloads the safety application
-	Reset SafeLOGIC	Restarts the SafeLOGIC controller

Table 9: Operating elements for the SafeLOGIC controller and Remote Control dialog box

9.2.3 LED status indicators on SafeIO modules

The LED status indicators of the safe I/O modules are used for initial diagnostics of the input and output channels and the operating state of the module. An error in the dual-channel evaluation can be indicated via the 00 and 0C LEDs, for example.
The LED status indicators indicate warnings and errors via unique blinking codes and permanent illumination.
For additional information, see the manual or Automation Help.

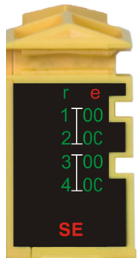


Figure 115: LED status indicators of a safe input module

Hardware \ X20 system \ X20 modules \ Analog input modules \ X20(c)SA4430 \ LED status indicators
Hardware \ X20 system \ X20 modules \ Digital output modules \ X20(c)SOx1x0 \ LED status indicators
Hardware \ X20 system \ X20 modules \ Digital input modules \ X20(c)SIx1x0 \ LED status indicators
Hardware \ X20 system \ X20 modules \ Temperature modules \ X20ST4492 \ LED status indicators
Hardware \ X20 system \ X20 modules \ Counter modules \ X20(c)SD1207 \ LED status indicators

9.3 Module replacement and update

9.3.1 Replacing a module

The system checks the safety-related hardware configuration at a time interval defined by the system. If a new module is detected, this is indicated via the SafeLOGIC controller.

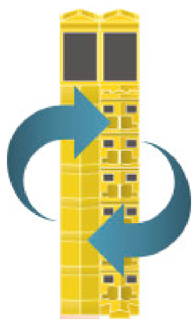


Figure 116: Replacing a module

Remote Control dialog box:

1) "(Number of new modules) Module exchange" is displayed as the module status of in the Remote Control dialog box.

2) Confirm with orange marked button 1, 2, 3, 4 or n.

3) Perform a test of the affected portion of the machine.

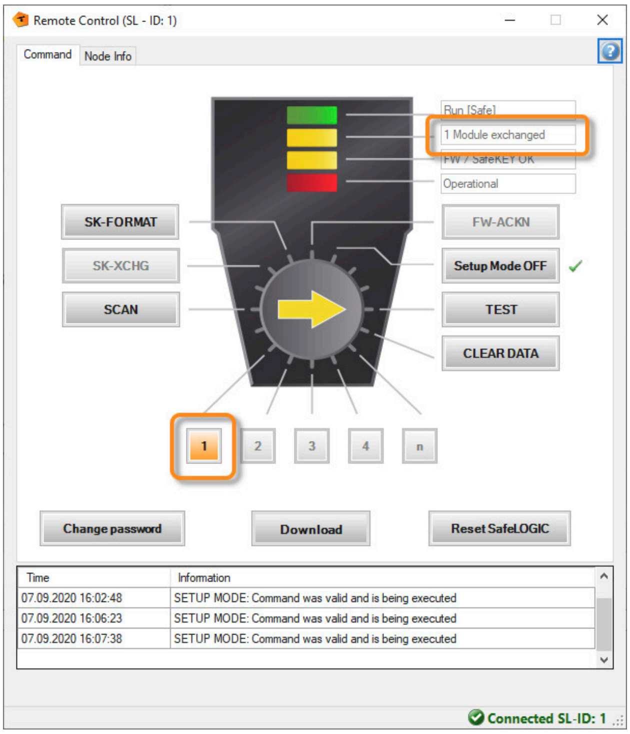


Figure 117: Confirming module replacement via the Remote Control dialog box

mapp Safety HMI application:

1) "Safenodes acknowledge required" is displayed in the "State" field in the top right corner.

2) Confirm with button "Acknowledge exchange" in the "Exchange" area.

3) Perform a test of the affected portion of the machine.

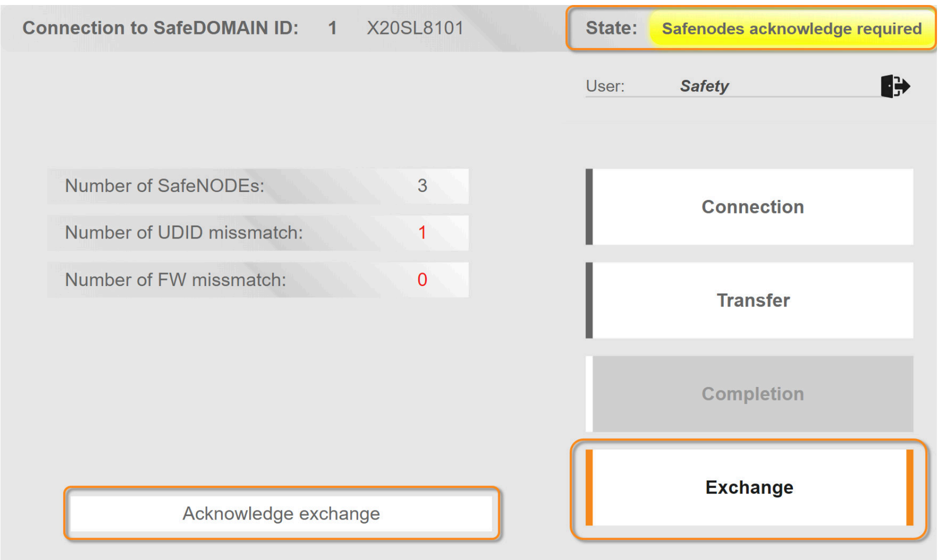


Figure 118: Confirming module replacement in the mapp Safety HMI application

> The Exchange dialog box is there to exchange SafeNODEs.
> A SafeLOGIC or SafeLOGIC-X controller exchange is therefore not possible.

SafeLOGIC:

1) LED MXCHG blinks (indicates the number of new modules)

2) Set 1, 2, 3, 4 or n for the selector switch and confirm with ENTER.

3) Perform a test of the affected portion of the machine.

> If more modules were replaced than indicated, a manual scan can be started via the SCAN button. This might be necessary for large machines on which the automatic scan takes more time.

9.3.2 Replacing the SafeLOGIC controller

A SafeLOGIC controller is replaced in the same way as a normal module. When replacing a SafeLOGIC controller, the SafeKEY from the SafeLOGIC controller being replaced must be applied in order to avoid enabling an old safety-related application.
The safety application and the device parameters of a SafeLOGIC-X controller are stored in the safety section of the flash memory of a controller. When the SafeLOGIC-X controller is replaced, the control system is automatically accessed and the current SafeAPPLICATION downloaded.

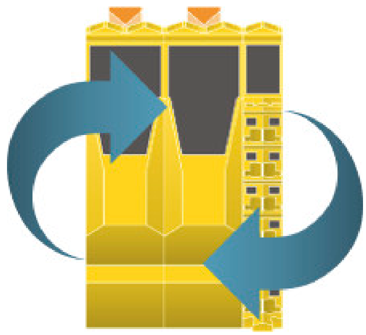


Figure 119: Replacing the SafeLOGIC controller

1) Replace the SafeLOGIC controller / SafeLOGIC-X controller.

2) Only with SafeLOGIC: Connect the old SafeKEY.

3) Acknowledge the new SafeKEY after booting:

   - Remote Control dialog box: Confirm with button SK-XCHG.
   - SafeLOGIC: Set SK-XCHG for the selector switch and confirm with ENTER.

4) The module to be acknowledged is displayed.

5) Acknowledge the new SafeLOGIC controller:

   - Remote Control dialog box: Confirm with orange marked button "1".
   - SafeLOGIC: Set "1" for the selector switch and confirm with ENTER.

6) No test is required.

> With new hardware, it may be that a firmware update is performed on the SafeLOGIC controller / SafeLOGIC-X controller. This extends the startup time and an entry is created in the Logger in Automation Studio.

> It is currently not yet possible to perform a SafeLOGIC or SafeLOGIC-X exchange via the mapp Safety HMI application.
> In this case, commissioning must be performed via the HMI application starting with a "SafeKEY format".

9.3.3 Confirming a firmware update

After a module has been replaced, the safety application has been extended by further modules or a firmware update has been installed, it may be necessary that the system performs a firmware update on a module.

Figure 120: Updating the firmware

Remote Control dialog box:

1) After a firmware update has been completed, "FW updated" is displayed in the system status of the Remote Control dialog box.
2) Acknowledge with orange marked button "FW-ACKN".
3) The modules are updated and started.
4) Perform a complete test of the safety application.

SafeLOGIC:

1) The FW-ACKN LED blinks after the firmware update.
2) Set FW-ACKN for the selector switch and confirm with ENTER.
3) The modules are updated and started.
4) Perform a complete test of the safety application.

9.3.4 Module is missing

The system checks the safety-related hardware configuration at a defined time interval. If a missing module is detected, this is indicated in the Remote Control dialog box, in System Diagnostics Manager and/or via blinking LED status indicators on the SafeLOGIC controller.

Figure 121: Module is missing

Remote Control dialog box:

- Displayed in the module status "(Number of missing modules) modules missing"

System Diagnostics Manager:

- Hardware error → Displayed in detail in the hardware tree

SafeLOGIC:

- LED MXCHG blinks fast and LED FAILSAFE blinks twice.

9.3.5 Updating the safety application

The safety application can be updated either via the mapp Safety HMI application, the Remote Control dialog box or by connecting a preprogrammed SafeKEY (only with SafeLOGIC).
For a download via the mapp Safety HMI application, it must be ensured that the most current file is available on the controller.
To ensure that the safe configuration and the safe parameters are not lost when the SafeKEY is replaced, the SK-COPY function of the SafeLOGIC controller can be used to copy the settings.

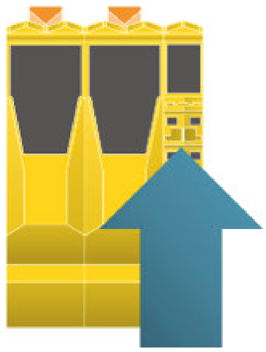


Figure 122: Update via the SafeKEY

mapp Safety HMI application:

1) Download the latest version of SafeAPPLICATION to the controller via Automation Studio.
2) Open the mapp Safety HMI application (http://<IP-ADRESSE>:81/index.html?visuId=mappSAFETY).
3) Log in and connect to the SafeLOGIC controller.
4) Select the SafeAPPLICATION in the "Transfer" area.
5) Update the SafeAPPLICATION via button "Transfer to SafeDOMAIN".
6) Perform a complete test of the safety application.

Remote Control dialog box:

1) Compile a SafeDESIGNER project.
2) Opening the Remote Control dialog box
3) Transfer the project via the "Download" button.
4) Perform a complete test of the safety application.

SafeLOGIC:

1) Old SafeKEY is connected.
2) Set SK-COPY for the selector switch and confirm with ENTER.
3) Configuration data from the SafeKEY are copied into the RAM of the SafeLOGIC controller.
4) Remove the old SafeKEY and connect the new one.
5) Confirm with ENTER; SafeLOGIC is rebooted.
6) Perform a complete test of the safety application.

> Safety technology \ mapp Safety \ Diagnostics and service \ Maintenance scenarios \ SafeKEY or safety section of the CompactFlash card \ Changing the application on the SafeLOGIC controller by replacing the SafeKEY (X20SL8xxx series only)

Exercise: Replace safe I/O module

The objective of this exercise is to disable module monitoring for the safe modules, to replace a safe module with another during operation and to acknowledge the module replacement.

1) Disable module monitoring in Automation Studio.
2) Compile the changes and transfer them to the controller.
3) Disconnect the SI module.
4) Connect the SI module with a different serial number.
5) Acknowledge module replacement on the SafeLOGIC controller or in the Remote Control dialog box.
6) Perform a test of the affected portion of the machine.

10 Further information

Further sources of information are available to deepen participants' understanding of the previous topics.

Topic	Source
B&R website \ Technology \ Safety technology • Safety technology • openSAFETY	www.br-automation.com/en-gb/technologies/safety-technology/
B&R website \ Products \ Safety technology	www.br-automation.com/en-gb/products/safety-technology/
mapp Safety in Automation Help Safety technology \ mapp Safety	 Figure 123: mapp Safety help documentation - Start page
Integrated safe motion control	Training module TM540

10.1 Project documentation and printing

SafeDESIGNER can be used to create the documentation of the safety application. The documentation interface can be opened via main menu "Project \ Project information".

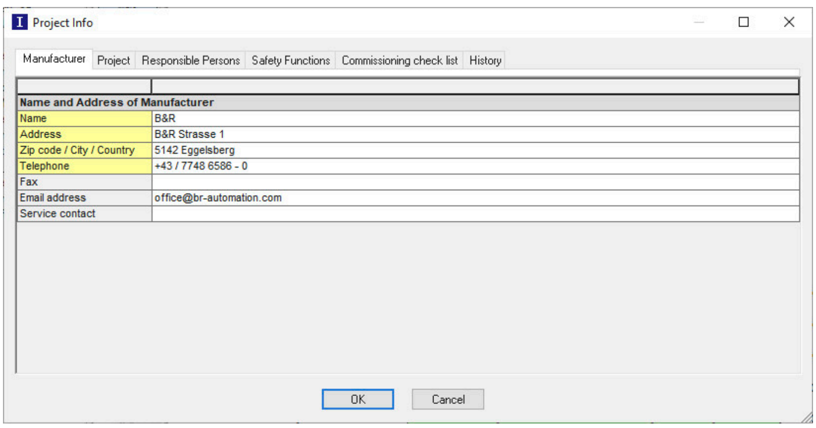


Figure 124: Project documentation

The following information belongs to the project information:

- Manufacturer
  Contact details of the machine manufacturer, such as name and address, are entered here.
- Project
  Data about the machine and the safety application are entered here. The project data is automatically maintained by the system and the unique CRC number of the safety application is also available.
- Persons in charge
  This tab is used to enter the persons who are in charge of the project, such as the project manager, the programmer of the safety application and the person who tests the safety application.
- Safety functions
  The safety functions configured for the machine are entered here. Information about a safety function that was successfully tested during integration can be entered here.
- Commissioning checklist
  This list serves as a support for validation. The programmer of the safety application enters data, for example for the network connections and wiring, here. This data must be observed later during commissioning.
- History
  The history of the safety application is managed here. For each revision, the corresponding CRC number and revision history is entered.

Yellow fields should always be filled in. Grey fields can be filled in optionally.

Project printout

After the project information has been filled in, the project documentation can be printed. The selection dialog box is displayed via main menu "File \ Printing project".
Additional settings for the documentation can be made using button "Page layout texts".

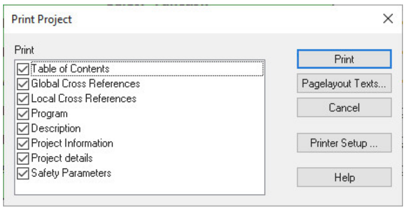


Figure 125: Print selection for the documentation

Safety technology \ mapp Safety \ Programming \ SafeDESIGNER \ Developing a project \
- Dialog box "Project info"
- Printing a project

10.2 How to connect to a safety controller

There are several ways to connect to a safety controller. The following table shows which of the connection types work with the various safety controllers. The following pages explain the connection options in more detail.

Connection options	SafeLOGIC	SafeLOGIC-X
Online connection via a standard CPU	✓	✓
Online via direct connection	✓	✗
Manual connection	✓	✗

Table 10: Options for connecting to the SafeLOGIC controller / SafeLOGIC-X controller

An online connection via a standard CPU is used in this training module.

Online connection via a standard CPU

The existing connection with the standard CPU can be used to establish online communication with the safety controller. The standard CPU automatically routes the data for this.

When SafeDESIGNER is opened, the current online settings of Automation Studio are called. If there is an online connection to a controller at this time, this IP address is used for the online settings in SafeDESIGNER and a connection can be made to the safety controller.

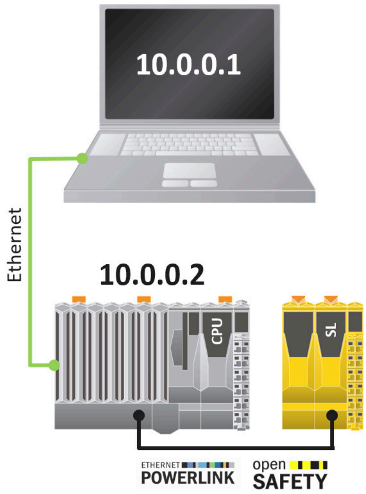


Figure 126: Online via a standard CPU

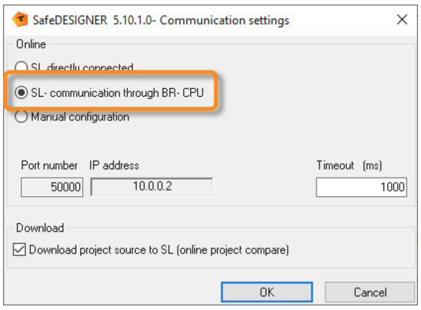


Figure 127: Communication settings "SL - communication through BR-CPU"

In SafeDESIGNER, communication to the SafeLOGIC controller can be configured via menu option **Online / Communication settings**. The preset default value is the connection "SL - communication through BR-CPU".

Online via direct connection

Using the direct connection, the PC is connected directly to the SafeLOGIC controller via a network cable. An IP address from the address range of the POWERLINK network (192.168.100.xxx) is set on the PC. The SafeLOGIC controller always uses the set node number as the last position of the IP address.

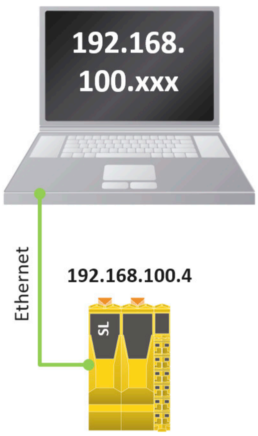


Figure 128: Online via direct connection

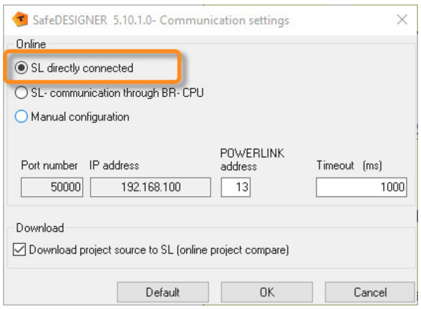


Figure 129: Communication settings "SL directly connected"

The dialog box for the connection settings in SafeDESIGNER is opened under "Online \ TCPIP communication parameters". Option "SL directly connected" must be selected for the direct connection.

Manual connection

When connecting manually, the user can set all communication parameters, e.g. the IP address and port number.

Safety technology \ mapp Safety \ Programming \ SafeDESIGNER \ Commissioning the safety controller \ Communication settings

10.3 Support for third-party devices with openSAFETY

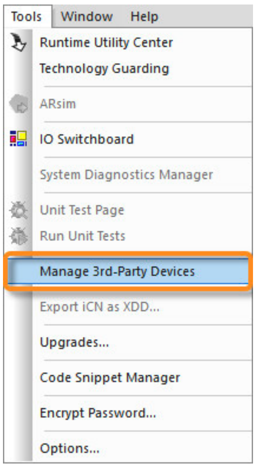


Figure 130: Adding 3rd-party devices

Operating an openSAFETY third-party device in Automation Studio requires an .xosdd and .xdd file from the device manufacturer. These files contain information about the configuration and functionality of the device.
New devices can be added via menu item "Extras \ Managing 3rd-party devices".

With "Importing fieldbus device(s)", a selection dialog box is opened in the current dialog box where the .xosdd file must be selected first and then the .xdd file.

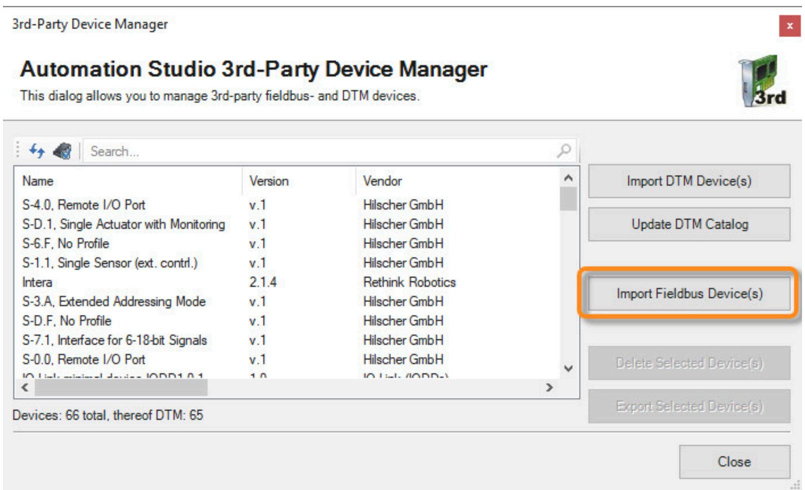


Figure 131: Selecting a fieldbus device

After the files have been imported, the device can be added and configured via the Hardware Catalog.

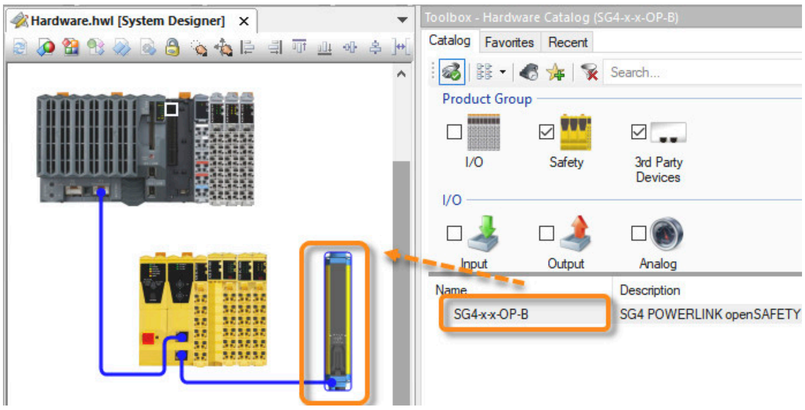


Figure 132: Adding a 3rd-party device

10.4 Example projects and solutions

The example code in this documentation is exclusively for implementing the test setup for this training. This example code is not permitted to be adopted for safety applications under any circumstances. Each safe machine application must be subjected to a separate risk analysis and a safety concept must be created.

10.4.1 Sample solution - Project

Main code

The main code contains the query for the "ModuleOk" parameters, the self-created function blocks and the connection between the reset signal and ReleaseOutput.

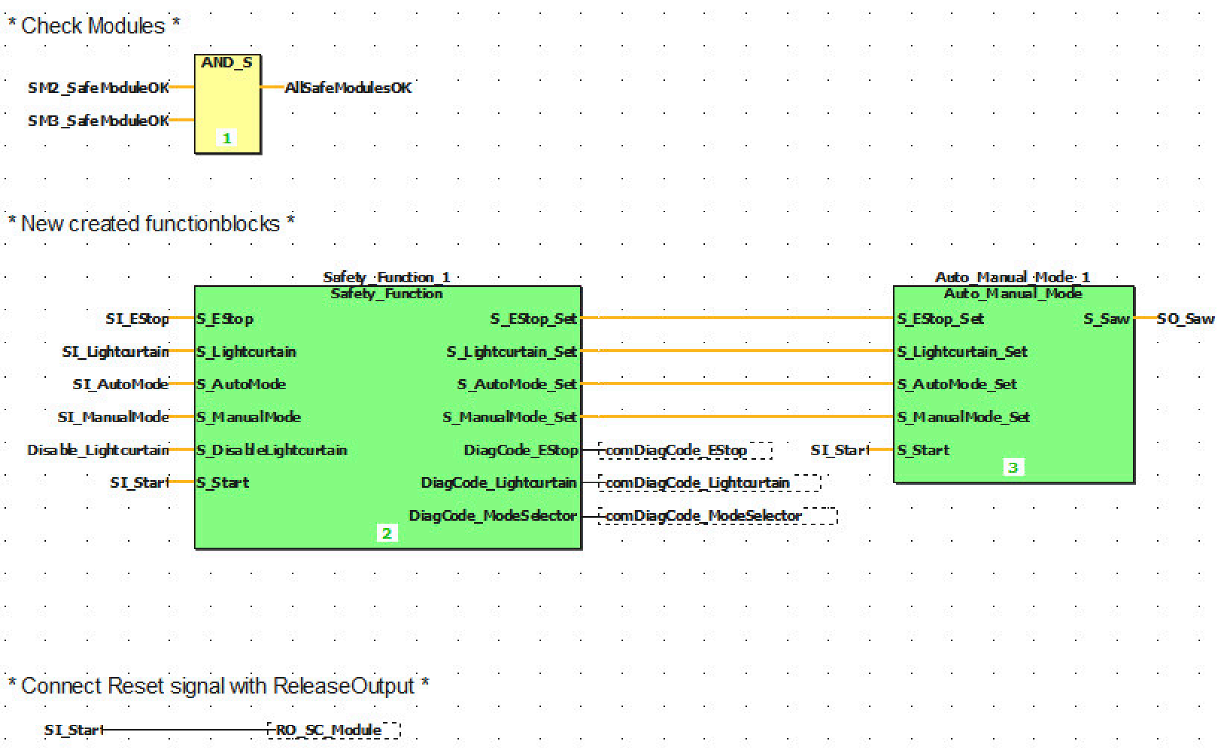


Figure 133: Code: Main

Function block: Auto_Manual_Mode

Functionality for automatic and manual mode is implemented in this function block.

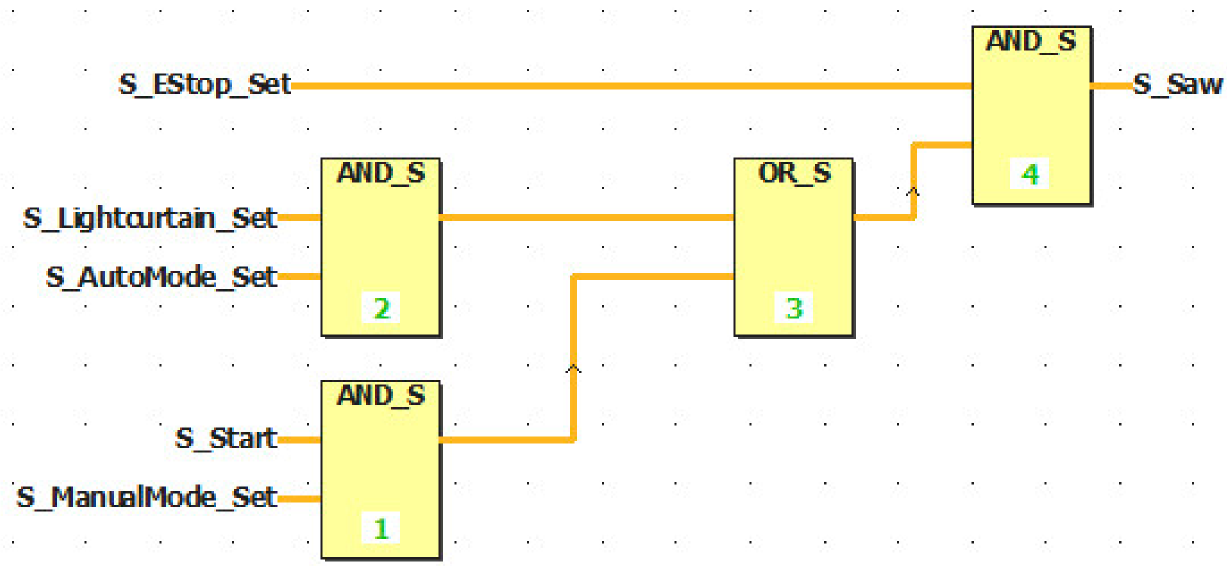


Figure 134: Auto_Manual_Mode function block

Function block: Safety_Function

The PLCopen function blocks for emergency stop, light curtain and mode selector switch are called and connected in this function block. Enabling and disabling the light curtain is also implemented via the SafeCOMMISSIONING file.

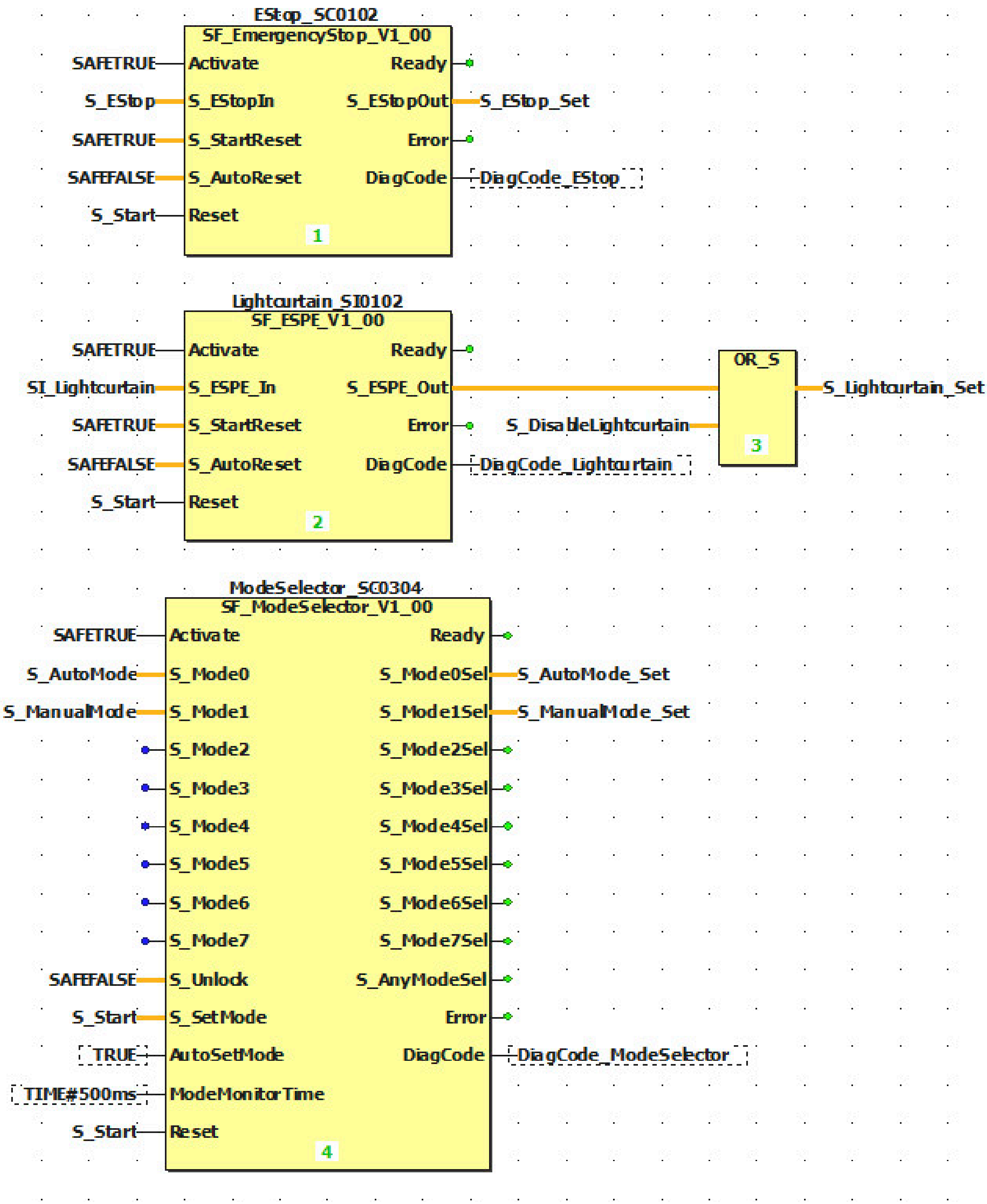


Figure 135: Safety_Function function block

10.4.2 Sample solution - Variables

Global variables

This file contains all variables that are connected to a safe input or output as well as the variables for the communication channels.

		Name	Data type	Description	Terminal
1	⊟	**ModuleOK**			
2		SM2_SafeModuleOK	SAFEBOOL		SL1.SM2.SafeModuleOK
3		SM3_SafeModuleOK	SAFEBOOL		SL1.SM3.SafeModuleOK
4		Disable_Lightcurtain	SAFEBOOL		SL1.SM1.SafeCommissioningOptionBIT000
5	⊟	**SafeInput**			
6		SI_EStop	SAFEBOOL		SL1.SM3.SafeTwoChannelInput0102
7		SI_Lightcurtain	SAFEBOOL		SL1.SM2.SafeTwoChannelInput0102
8		SI_AutoMode	SAFEBOOL		SL1.SM3.SafeDigitalInput03
9		SI_ManualMode	SAFEBOOL		SL1.SM3.SafeDigitalInput04
10		SI_Start	SAFEBOOL		SL1.SM3.SafeDigitalInput05
11	⊟	**SafeOutput**			
12		SO_Saw	SAFEBOOL		SL1.SM3.SafeDigitalOutput01
13	⊟	**ReleaseOutput**			
14		RO_SC_Module	BOOL		SL1.SM3.ReleaseOutput
15	⊟	**Communication**			
16		comDiagCode_EStop	WORD		SL1.SM1.UINT0001
17		comDiagCode_Lightcurtain	WORD		SL1.SM1.UINT0002
18		comDiagCode_ModeSelector	WORD		SL1.SM1.UINT0003

Figure 136: Global variable declaration

Local variables: Main code

Created user function blocks and the variable "AllSafeModulesOK" for the module status of all safe modules are here.

	Name	Data type	Usage	Description	Init	Diag	Feedback
1	⊟ **FB**						
2	AllSafeModulesOK	SAFEBOOL	VAR				☐
3	Auto_Manual_Mode_1	Auto_Manual_...	VAR				☐
4	Safety_Function_1	Safety_Function	VAR				☐

Figure 137: Local variable declaration for main code

Local variables: SafetyFunction

The inputs on this function block serve as an interface for the safe inputs and for the reset signal. The instances of the PLCopen function blocks are declared as internal variables and the status of the safety functions and the diagnostic codes of the PLCopen function blocks are declared as outputs.

	Name	Data type	Usage	Description	Init	Diag	Feedback
1	⊟ **Input**						
2	S_EStop	SAFEBOOL	VAR_INPUT				☐
3	S_Lightcurtain	SAFEBOOL	VAR_INPUT				☐
4	S_AutoMode	SAFEBOOL	VAR_INPUT				☐
5	S_ManualMode	SAFEBOOL	VAR_INPUT				☐
6	S_DisableLightcurtain	SAFEBOOL	VAR_INPUT				☐
7	S_Start	SAFEBOOL	VAR_INPUT				☐
8	⊟ **Internal Variables**						
9	EStop_SC0102	SF_EmergencyStop_V1_00	VAR				☐
10	Lightcurtain_SI0102	SF_ESPE_V1_00	VAR				☐
11	ModeSelector_SC0304	SF_ModeSelector_V1_00	VAR				☐
12	⊟ **Output**						
13	S_EStop_Set	SAFEBOOL	VAR_OUTPUT			☐	☐
14	S_Lightcurtain_Set	SAFEBOOL	VAR_OUTPUT			☐	☐
15	S_AutoMode_Set	SAFEBOOL	VAR_OUTPUT			☐	☐
16	S_ManualMode_Set	SAFEBOOL	VAR_OUTPUT			☐	☐
17	DiagCode_EStop	WORD	VAR_OUTPUT			☐	☐
18	DiagCode_Lightcurtain	WORD	VAR_OUTPUT			☐	☐
19	DiagCode_ModeSelector	WORD	VAR_OUTPUT			☐	☐

Figure 138: Local variable declaration for SafetyFunction function block

Local variables: Auto_Manual_Mode

In the Auto_Manual_Mode function block, the information of the PLCopen safety functions (emergency stop, light curtain and mode selector switch) and the green button are declared as inputs and the signal for the safe output as output.

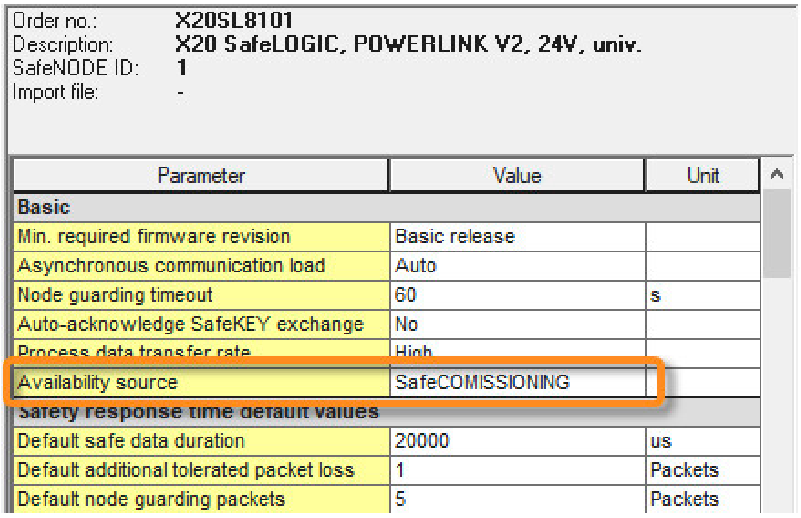
	Name	Data type	Usage	Description	Init	Diag	Feedback
1	⊟ **Input**						
2	S_EStop_Set	SAFEBOOL	VAR_INPUT				☐
3	S_Lightcurtain_Set	SAFEBOOL	VAR_INPUT				☐
4	S_AutoMode	SAFEBOOL	VAR_INPUT				☐
5	S_ManualMode	SAFEBOOL	VAR_INPUT				☐
6	S_Start	SAFEBOOL	VAR_INPUT				☐
7	⊟ **Output**						
8	S_Saw	SAFEBOOL	VAR_OUTPUT			☐	☐

Figure 139: Local variable declaration for the Auto_Manual_Mode function block

10.4.3 Sample solution - Parameters for the safe hardware modules

Parameter SafeLOGIC

With the SafeLOGIC controller, "SafeCOMMISSIONING" only needs to be set as "Availability Source".

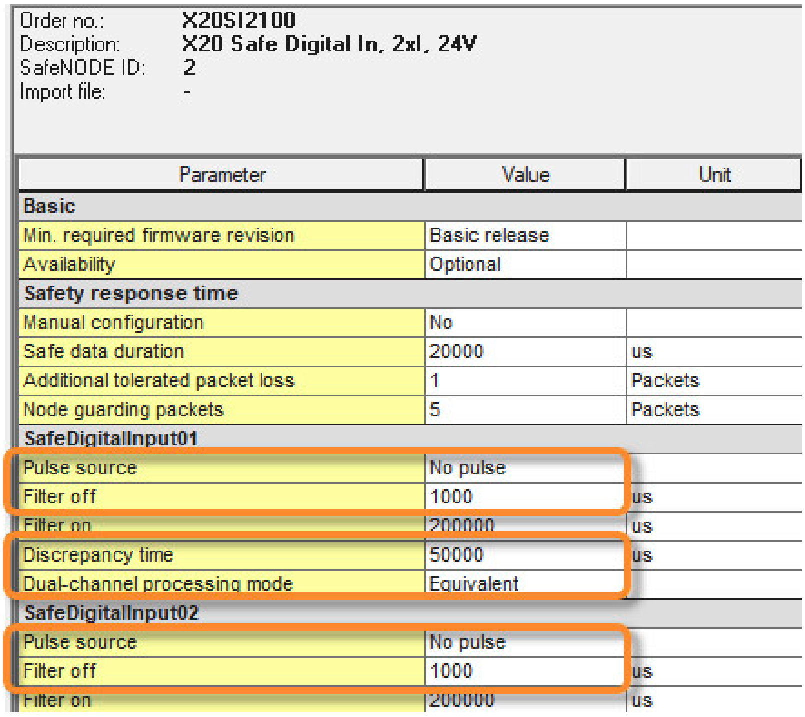

Order no.: X20SL8101
Description: X20 SafeLOGIC, POWERLINK V2, 24V, univ.
SafeNODE ID: 1
Import file: -

Parameter	Value	Unit
Basic		
Min. required firmware revision	Basic release	
Asynchronous communication load	Auto	
Node guarding timeout	60	s
Auto-acknowledge SafeKEY exchange	No	
Process data transfer rate	High	
Availability source	SafeCOMISSIONING	
Safety response time default values		
Default safe data duration	20000	us
Default additional tolerated packet loss	1	Packets
Default node guarding packets	5	Packets

Figure 140: Parameter X20SL8101

Parameter X20SI2100

The inputs from the safe input module are wired to the light curtain. This sensor already has its own test pulse, so it has to be disabled on the module and a filter must be configured. The light curtain is evaluated via multi-channel evaluation using an equivalent signal.

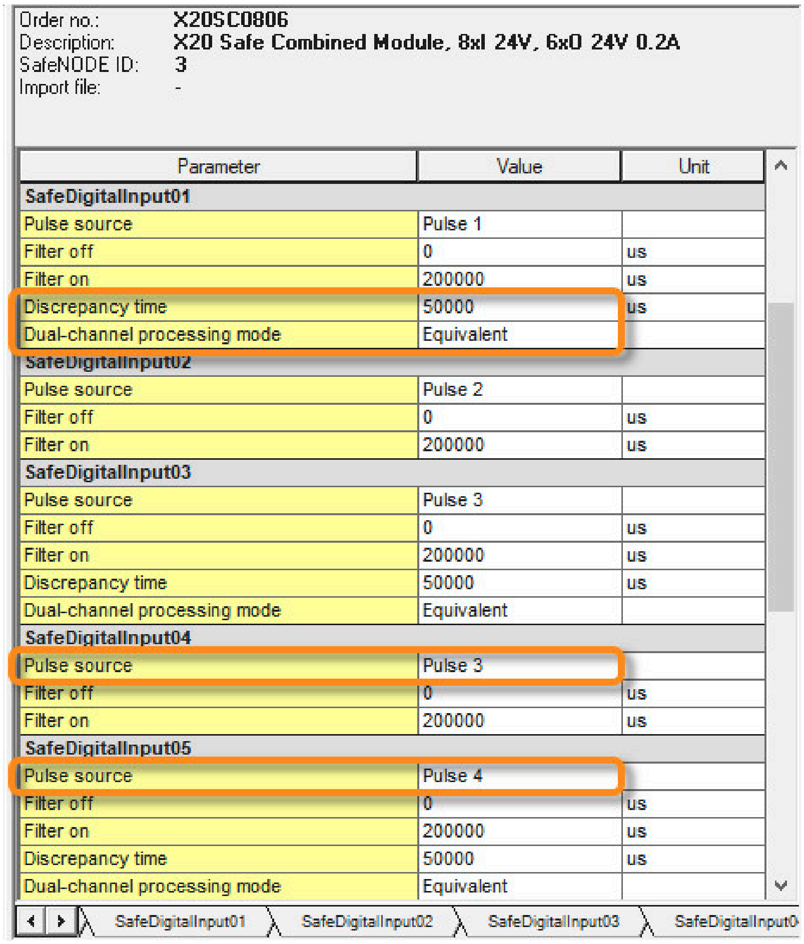

Order no.: X20SI2100
Description: X20 Safe Digital In, 2xI, 24V
SafeNODE ID: 2
Import file: -

Parameter	Value	Unit
Basic		
Min. required firmware revision	Basic release	
Availability	Optional	
Safety response time		
Manual configuration	No	
Safe data duration	20000	us
Additional tolerated packet loss	1	Packets
Node guarding packets	5	Packets
SafeDigitalInput01		
Pulse source	No pulse	
Filter off	1000	us
Filter on	200000	us
Discrepancy time	50000	us
Dual-channel processing mode	Equivalent	
SafeDigitalInput02		
Pulse source	No pulse	
Filter off	1000	us
Filter on	200000	us

Figure 141: Parameter X20SI2100

Parameter X20SC0806

The safe mixed module is wired to the emergency stop, the mode selector switch, the green button and the "Saw" output. The emergency stop signal is read out equivalently. The pulse for the mode selector switch and green button must be changed.



Order no.: X20SC0806
Description: X20 Safe Combined Module, 8xI 24V, 6xO 24V 0.2A
SafeNODE ID: 3
Import file: -

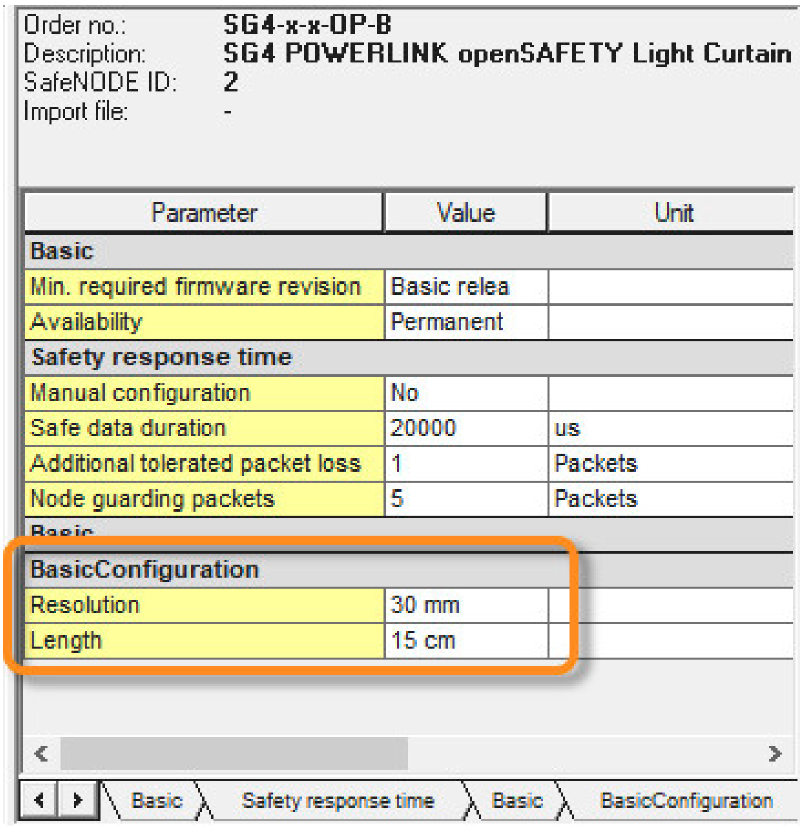
Parameter	Value	Unit
SafeDigitalInput01		
Pulse source	Pulse 1	
Filter off	0	us
Filter on	200000	us
Discrepancy time	50000	us
Dual-channel processing mode	Equivalent	
SafeDigitalInput02		
Pulse source	Pulse 2	
Filter off	0	us
Filter on	200000	us
SafeDigitalInput03		
Pulse source	Pulse 3	
Filter off	0	us
Filter on	200000	us
Discrepancy time	50000	us
Dual-channel processing mode	Equivalent	
SafeDigitalInput04		
Pulse source	Pulse 3	
Filter off	0	us
Filter on	200000	us
SafeDigitalInput05		
Pulse source	Pulse 4	
Filter off	0	us
Filter on	200000	us
Discrepancy time	50000	us
Dual-channel processing mode	Equivalent	

SafeDigitalInput01 | SafeDigitalInput02 | SafeDigitalInput03 | SafeDigitalInput0

Figure 142: Parameter X20SC0806

Parameter openSAFETY Datalogic light curtain

With the Datalogic light curtain, the parameters for resolution and length must be changed depending on the model type.



Order no.: SG4-x-x-OP-B
Description: SG4 POWERLINK openSAFETY Light Curtain
SafeNODE ID: 2
Import file: -

Parameter	Value	Unit
Basic		
Min. required firmware revision	Basic relea	
Availability	Permanent	
Safety response time		
Manual configuration	No	
Safe data duration	20000	us
Additional tolerated packet loss	1	Packets
Node guarding packets	5	Packets
Basic		
BasicConfiguration		
Resolution	30 mm	
Length	15 cm	

Basic | Safety response time | Basic | BasicConfiguration

Figure 143: Parameter SG4-x-x-OP-B

11 Summary

The first steps for a safety application are done in Automation Studio by configuring all safe hardware components and mapp Safety. After this has been completed, the next step is to start the SafeAPPLICATION in SafeDESIGNER.
SafeDESIGNER is used for the safety-related configuration of the safety controller and safe modules. The safety application can be configured using the visual editor and a large number of PLCopen Safety function blocks. A convenient user interface and numerous diagnostic options facilitate commissioning of the safety controller.
The mapp Safety HMI application makes it possible to download a SafeAPPLICATION and select the SafeCOMMISSIONING file even without SafeDESIGNER. In addition to the advantages already mentioned, SafeDESIGNER provides complete project documentation and an integrated simulation environment.

Figure 144: mapp Safety

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